

# Erdős-Hajnal conjecture beyond five-vertex graphs

Shenwei Huang<sup>1</sup>, Yiao Ju<sup>2</sup>, and Yidong Zhou<sup>2</sup>

<sup>1</sup>School of Mathematical Sciences and LPMC, Nankai University, Tianjin 300071, P.R. China. Email: [shenweihuang@nankai.edu.cn](mailto:shenweihuang@nankai.edu.cn).

<sup>2</sup>College of Computer Science, Nankai University, Tianjin 300350, China.

September 1, 2026

## Abstract

In 1989, Erdős and Hajnal conjectured that for any graph  $H$ , there is a constant  $c = c(H) > 0$  such that every  $n$ -vertex graph  $G$  with no induced copies of  $H$  contains a clique or an independent set of size at least  $n^c$ . This conjecture, known as the Erdős-Hajnal conjecture, is a central open problem in combinatorics and listed as one of the top 10 Erdős problems by Bloom on the Erdős problem website <https://www.erdosproblems.com/>. In a recent breakthrough, Nguyen, Scott and Seymour proved that Erdős-Hajnal conjecture holds for the case when  $H$  is the five-vertex path, which, combined with known results, implies that Erdős-Hajnal conjecture holds for every five-vertex graph.

In this paper, we extend the iterative sparsification framework recently developed by Nguyen, Scott and Seymour. We introduce a generalized niceness condition relaxing their nice condition, a novel intermediate property concerning combs and a general structural lemma (which may be of independent interest) that is sufficient to deduce the Erdős-Hajnal conjecture.

This framework simultaneously recovers the recent result on the five-vertex path (PLMS 2026) and the classical result on the bull graph by Chudnovsky and Safra (JCTB 2008) as special cases, thereby unifying these two previously independent strands, and further proves the conjecture for two new cases: the E-graph (which contains the five-vertex path) and the Bird graph (which contains both the five-vertex path and the bull). These are the first two six-vertex graphs whose validity does not follow from the known operations (see Alon-Pach-Solymosi, Combinatorica 2001, and Nguyen-Scott-Seymour, TAMS 2026) that preserve the Erdős-Hajnal property.

## 1 Introduction

A *clique* (*stable set*) is a vertex set whose vertices are pairwise adjacent (nonadjacent, resp.). Ramsey [17] proved that for every integer  $k \geq 1$ , there exists a positive integer  $R(k, k)$  such that every graph on at least  $R(k, k)$  vertices contains a clique or stable set of size  $k$ . This shows that a sufficiently large graph must contain a large clique or stable set. Erdős [9] proved via probabilistic method that for large  $n$ , there exists an  $n$ -vertex graph with no cliques or stable

sets of size at least  $2 \log n$ . This shows that general graphs do not necessarily contain a clique or stable set of polynomial size. However,  $H$ -free graphs might behave differently.

Let  $G, H$  be graphs. We say that  $G$  is  $H$ -free if  $G$  has no induced subgraph that is isomorphic to  $H$ . In 1989, Erdős and Hajnal [10] showed that for any graph  $H$ , there exists a constant  $c = c(H) > 0$  such that any  $n$ -vertex  $H$ -free graph has a clique or stable set of size at least  $e^{c\sqrt{\log n}}$ . Erdős and Hajnal [10] further conjectured that one can improve the bound to a polynomial of  $n$ , which has become known as the Erdős-Hajnal conjecture.

**Conjecture 1.** (Erdős-Hajnal [10]) *For every graph  $H$ , there exists  $c = c(H) > 0$  such that every  $n$ -vertex  $H$ -free graph has a clique or stable set of size at least  $n^c$ .*

In 2024, Bucić, Nguyen, Scott and Seymour [5] gave an  $\log \log$  improvement over the classical Erdős-Hajnal bound: for any graph  $H$ , there exists a constant  $c = c(H) > 0$  such that any  $n$ -vertex  $H$ -free graph has a clique or stable set of size at least  $e^{c\sqrt{\log n \log \log n}}$ . For a set  $\mathcal{H}$  of graphs,  $G$  is  $\mathcal{H}$ -free if  $G$  is  $H$ -free for all  $H \in \mathcal{H}$ . The Erdős-Hajnal conjecture is equivalent to the statement that for any nonempty set  $\mathcal{H}$  of graphs, there exists a constant  $c = c(\mathcal{H}) > 0$  such that any  $n$ -vertex  $\mathcal{H}$ -free graph has a clique or stable set of size  $n^c$ . If a nonempty set  $\mathcal{H}$  of graphs satisfies this statement, we say that  $\mathcal{H}$  has the *Erdős-Hajnal property*. In this case, We call  $c(\mathcal{H})$  a *Erdős-Hajnal constant* for  $\mathcal{H}$ . It is clear that a graph  $H$  has the Erdős-Hajnal property if and only if  $\overline{H}$  has the Erdős-Hajnal property, where  $\overline{H}$  is the complement of  $H$ .

## 1.1 Equivalent formulations

The Rödl theorem says that every graph in a proper hereditary graph class contains an induced subgraph with linear size that is very sparse or very dense. Formally, we say that  $X$  is  $c$ -sparse if every vertex in  $X$  has at most  $c|X|$  neighbors in  $X$ , and  $X$  is  $c$ -restricted if  $X$  or  $\overline{X}$  is  $c$ -sparse.

**Theorem 1.1.** (Rödl [18]) *For every graph  $H$  and every  $\epsilon \in (0, \frac{1}{2})$ , there exists  $\delta > 0$  such that every  $H$ -free graph  $G$  has an  $\epsilon$ -restricted induced subgraph of size at least  $\delta|G|$ .*

The original proof of Rödl uses the regularity lemma which gave a tower-type bound on  $\delta$ . Fox and Sudakov [12] made a significant improvement showing that one can take  $\delta = e^{-c|H|(\log \frac{1}{\epsilon})^2}$  for some constant  $c > 0$ , and they conjectured that one can take  $\delta$  to be a polynomial in  $\epsilon$ . A set  $\mathcal{H}$  of graphs (or the class of  $\mathcal{H}$ -free graphs) has the *polynomial Rödl property*, if there exists  $d \geq 1$  such that for every  $\epsilon \in (0, \frac{1}{2})$ , every  $\mathcal{H}$ -free graph  $G$  has an  $\epsilon$ -restricted induced subgraph of size at least  $\epsilon^d|G|$ . By taking  $\epsilon$  to be a small polynomial of  $|G|^{-1}$ , one may see that the polynomial Rödl property implies the Erdős-Hajnal property.

For two graphs  $G$  and  $H$ , an induced *copy* of  $H$  in  $G$ , is an injection  $\phi : V(H) \rightarrow V(G)$  such that for all  $u, v \in V(H)$ ,  $uv \in E(H)$  if and only if  $\phi(u)\phi(v) \in E(G)$ . The number of induced copies of  $H$  in  $G$  is denoted by  $\text{ind}_H(G)$ . Nikiforov generalized Theorem 1.1 to the graphs that contain few induced copies of  $H$ .

**Theorem 1.2.** (Nikiforov [16]) *For every graph  $H$  and every  $\epsilon \in (0, \frac{1}{2})$ , there exists  $\delta > 0$  such that every graph  $G$  with  $\text{ind}_H(G) < (\delta|G|)^{|H|}$  has an  $\epsilon$ -restricted induced subgraph of size at least  $\delta|G|$ .*

A finite set  $\mathcal{H}$  of graphs (or the class of  $\mathcal{H}$ -free graphs) is *viral*, if there exists  $d \geq 1$  such that every graph  $G$  with  $\text{ind}_H(G) < (\epsilon^d |G|)^{|H|}$  for all  $H \in \mathcal{H}$  has an  $\epsilon$ -restricted induced subgraph of size at least  $\epsilon^d |G|$ . Bucić, Fox and Pham [4] proved the equivalence of the Erdős-Hajnal property, the polynomial Rödl property, and the viral property.

**Theorem 1.3.** (Bucić-Fox-Pham [4]) *For a finite set  $\mathcal{H}$  of graphs,  $\mathcal{H}$  has the Erdős-Hajnal property if and only if it has the polynomial Rödl property if and only if it is viral.*

## 1.2 Erdős-Hajnal for a single graph

A vertex set  $S \subseteq V(G)$  is *homogeneous* in  $G$  if all vertices in  $S$  have the same neighborhood in  $V(G) \setminus S$ . A homogeneous set  $S$  of  $G$  is *nontrivial* if  $2 \leq |S| \leq |G| - 1$ . A graph  $G$  is *prime* if  $G$  has no nontrivial homogeneous set.

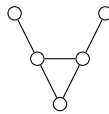
Let  $G_1$  and  $G_2$  be graphs with disjoint vertex sets, and  $a \in V(G_1)$ . We say that  $G$  is obtained from  $G_1$  by *substituting*  $G_2$  for  $a$  if:

- $V(G) = (V(G_1) \setminus \{a\}) \cup V(G_2)$ ,
- $u, v \in V(G_1) \setminus \{a\}$  are adjacent in  $G$  if and only if  $u, v$  are adjacent in  $G_1$ ,
- $u, v \in V(G_2)$  are adjacent in  $G$  if and only if  $u, v$  are adjacent in  $G_2$ ,
- $u \in V(G_1) \setminus \{a\}$  and  $v \in V(G_2)$  are adjacent in  $G$  if and only if  $u$  is adjacent to  $a$  in  $G_1$ .

It is clear that a graph  $G$  on at least two vertices is prime if and only if there do not exist graphs  $G_1, G_2$  with  $|G_1|, |G_2| \geq 2$  such that  $G$  is obtained from  $G_1$  by substituting  $G_2$  for a vertex in  $G_1$ .

One important result due to Alon, Pach and Solymosi [1] says that it suffices to consider Conjecture 1 for prime graphs.

**Theorem 1.4.** (Alon-Pach-Solymosi [1]) *Let  $H_1, H_2$  be graphs, and  $H$  be the graph obtained from  $H_1$  by substituting  $H_2$  for a vertex  $v$  in  $H_1$ . If both  $H_1, H_2$  have the Erdős-Hajnal property, then  $H$  also has the Erdős-Hajnal property.*



**Fig. 1.** The bull graph.

One can easily show via Theorem 1.4 that every graph with at most four vertices satisfies the Erdős-Hajnal property. There are four prime 5-vertex graphs: the bull (see Figure 1),  $C_5$  (the five-vertex cycle),  $P_5$  (the five-vertex path), and  $\overline{P_5}$ . In 2008, Chudnovsky and Safra [6] showed that the bull graph has the Erdős-Hajnal property. In 2023, Chudnovsky, Scott, Seymour and Spirkl [8] showed that

**Theorem 1.5** (Chudnovsky-Scott-Seymour-Spirkl [8]).  *$C_5$  has the Erdős-Hajnal property.*

Also in 2023, Nguyen, Scott and Seymour [15] showed that

**Theorem 1.6** (Nguyen-Scott-Seymour [15]).  $P_5$  and  $\overline{P_5}$  have the Erdős-Hajnal property.

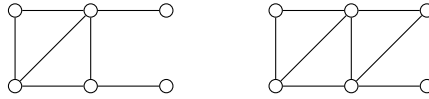
These results imply that all graphs on at most 5 vertices have the Erdős-Hajnal property. On the other hand, Nguyen, Scott and Seymour [13] constructed an infinitely family of prime graphs that have the Erdős-Hajnal property. The main insight from [13] is an operation that preserves the viral property.

**Theorem 1.7.** (Nguyen-Scott-Seymour [13]) *Let  $\mathcal{F}$  be a finite set of graphs, and let  $F_1, \overline{F_2} \in \mathcal{F}$ . For  $i = 1, 2$ , let  $v_i$  be a vertex of  $F_i$  with degree one, and let  $F'_i = F_i \setminus \{v_i\}$ . If  $\mathcal{F}_1 = \{F'_1\} \cup (\mathcal{F} \setminus \{F_1\})$  and  $\mathcal{F}_2 = \{\overline{F'_2}\} \cup (\mathcal{F} \setminus \{\overline{F_2}\})$  are both viral, then  $\mathcal{F}$  is viral.*

By combining Theorems 1.3 and 1.7, we have

**Corollary 1.8.** *Let  $\mathcal{F}$  be a finite set of graphs, and let  $F_1, \overline{F_2} \in \mathcal{F}$ . For  $i = 1, 2$ , let  $v_i$  be a vertex of  $F_i$  with degree one, and let  $F'_i = F_i \setminus \{v_i\}$ . If  $\mathcal{F}_1 = \{F'_1\} \cup (\mathcal{F} \setminus \{F_1\})$  and  $\mathcal{F}_2 = \{\overline{F'_2}\} \cup (\mathcal{F} \setminus \{\overline{F_2}\})$  have the Erdős-Hajnal property, then  $\mathcal{F}$  has the Erdős-Hajnal property.*

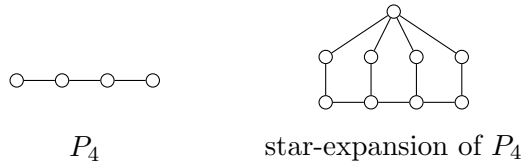
As a corollary, the two 6-vertex graphs in Figure 2 both have the Erdős-Hajnal property [13].



**Fig. 2.** Two prime 6-vertex graphs that have the Erdős-Hajnal property.

### 1.3 Erdős-Hajnal for two graphs

In the following we mention some results on Erdős-Hajnal property of graphs with more than one forbidden induced subgraph. Bousquet, Lagoutte and Thomassé [3] proved that for every integer  $k \geq 1$ ,  $\{P_k, \overline{P_k}\}$  has the Erdős-Hajnal property. A stronger result that if  $H_1, H_2$  are forests, then  $\{H_1, \overline{H_2}\}$  has the Erdős-Hajnal property [7] was proved by Chudnovsky, Scott, Seymour and Spirk. In the paper which proved that  $C_5$  has the Erdős-Hajnal property [8], the authors gave a generalized result. Let  $H$  be a graph with vertex set  $\{b_1, \dots, b_k\}$ . The *star-expansion* of  $H$ , is the graph obtained from  $H$  that adds  $k + 1$  vertices  $a_1, \dots, a_k, v$ , where  $a_i$  is adjacent to  $b_i$  for  $i \in [k]$ , and  $v$  is adjacent to  $a_1, \dots, a_k$ , and there are no other edges (see Figure 3 for an example). They proved the following theorem.

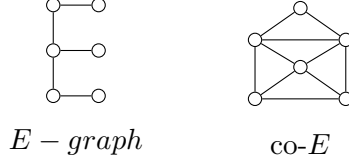


**Fig. 3.**  $P_4$  and its star-expansion.

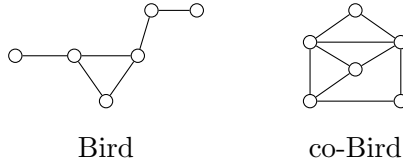
**Theorem 1.9.** (Chudnovsky-Scott-Seymour-Spirkl [8]) *Let  $H$  be a forest. Let  $H_1$  be the star expansion of  $H$ , and  $H_2$  be the star-expansion of  $\overline{H}$ . Then  $\{H_1, H_2, \overline{H_1}, \overline{H_2}\}$  has the Erdős-Hajnal property.*

Since  $P_4$  is the complement of itself and the star-expansion of  $P_4$  contains  $C_6, C_7$ , it follows that  $\{C_6, \overline{C_6}\}$  and  $\{C_7, \overline{C_7}\}$  have the Erdős-Hajnal property. In the same paper, it was proved that  $\{\hat{C}_5, \overline{\hat{C}_5}\}$  has the Erdős-Hajnal property, where  $\hat{C}_5$  is the graph obtained from  $C_5$  by adding a vertex that is adjacent to two adjacent vertices of the  $C_5$ .

#### 1.4 Our Contributions



**Fig. 4.** The graphs  $E$ -graph and its complement  $\text{co-}E$ .



**Fig. 5.** The graph Bird and its complement  $\text{co-Bird}$ .

Let  $E$ -graph and Bird be the graphs in Figures 4 and 5, respectively. In this paper, we prove the following theorems.

**Theorem 1.10.**  *$E$ -graph has the Erdős-Hajnal property.*

**Theorem 1.11.** *Bird has the Erdős-Hajnal property.*

Since both  $E$ -graph and Bird contain an induced  $P_5$ , both Theorem 1.10 and Theorem 1.11 generalize the main result by Nguyen, Scott and Seymour [15]. Since Bird graph contains the bull graph, Theorem 1.11 generalizes the result of Chudnovsky and Safra [6]. In fact, as we shall see, these two previously independent results are unified within our extended iterative sparsification framework (see the remark at the end of Section 5).

While the extension from five-vertex graphs to six-vertex graphs may appear incremental, the technical obstacles are substantial: the existing framework of Nguyen, Scott and Seymour does not directly apply to these cases, and new structural tools—a generalized niceness condition and a comb-based criterion—are required.

Our proofs are inspired by [15]. The proof of Theorems 1.10 and 1.11 proceeds in four main steps. First, we extend the framework of Nguyen, Scott and Seymour [15] by introducing a generalized niceness condition that relaxes the original notion of “nice”, and show that any finite class of graphs satisfying this property, along with two technical conditions called leaf-reducibility and wonderfulness (see Section 2 for precise definitions), has the Erdős-Hajnal property (see Section 3).

**Lemma 1.12.** *Let  $\mathcal{F}$  be a finite class of graphs that is leaf-reducible and wonderful. If  $\mathcal{F}$  is generalized nice, then  $\mathcal{F}$  has the Erdős-Hajnal property.*

Second, we introduce a new structural property, denoted by  $(*)$ , which concerns the existence of certain structures from a comb, and prove that under leaf-reducibility condition,  $(*)$  implies generalized niceness (see Section 4).

**Lemma 1.13.** *Let  $\mathcal{F}$  be a finite class of graphs that is leaf-reducible. If  $\mathcal{F}$  has property  $(*)$ , then  $\mathcal{F}$  is generalized nice.*

Third, we establish a general structural lemma that provides sufficient conditions for deducing  $(*)$  (see Section 5).

**Lemma 1.14.** *Let  $\mathcal{F}_1$  and  $\mathcal{F}_2$  be two finite sets of graphs that satisfy the Erdős-Hajnal property. Let  $\mathcal{H}$  be a finite set of graphs. Suppose that for every  $\overline{\mathcal{H}}$ -free graph  $G$  and every  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$  with  $\ell, w \geq 4$ , if there exists a special vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ ,  $B_i$  (for each  $i \in [\ell]$ ) can be partitioned into  $X_i, Y_i$  such that*

- (1)  $Y_i$  is  $\mathcal{F}_1$ -free;
- (2)  $X_i$  can be partitioned into  $(A_1^i, \dots, A_{t_i}^i)$  such that
  - (2.1)  $(A_1^i, \dots, A_{t_i}^i)$  is a pure blockade;
  - (2.2) the pattern of  $(A_1^i, \dots, A_{t_i}^i)$ , whose vertex set consists of all blocks of  $(A_1^i, \dots, A_{t_i}^i)$  and two vertices are adjacent if and only if their corresponding blocks are complete to each other, is  $\mathcal{F}_2$ -free;
  - (2.3) for each  $j \in [t_i]$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $A_j^i$ .

Then  $\mathcal{H}$  satisfies property  $(*)$ .

We make essential use of the comb structure, which was introduced by Chudnovsky, Scott, Seymour and Spirkel [8]. While the relevance of combs to the Erdős-Hajnal conjecture has been noted previously, to the best of our knowledge, this is the first time that a systematic framework-encompassing property  $(*)$ , generalized niceness, and two new graph classes has been built around them. In particular, our Lemma 1.14 provides a general structural criterion that goes beyond any individual application of combs in the literature.

Finally, we show that  $E$ -graph and Bird satisfy the hypothesis of Lemma 1.14 (see Section 6). One key step in the proof is to define certain equivalence relation on the set of blocks in a blockade and take quotient to obtain a new blockade whose pattern graph lies in a graph class which can be proved to have the Erdős-Hajnal property.

## 2 Preliminaries

All graphs in this paper are finite and simple. We follow [2] for general graph theory terminology that is not defined here. Let  $P_n, C_n$  and  $K_n$  denote the induced path, cycle and complete graph on  $n$  vertices, respectively. For positive integers  $s, t \geq 1$ , let  $K_{s,t}$  be the complete bipartite graph with one side having  $s$  vertices and other side having  $t$  vertices. If  $s = 1$ , the graph  $K_{1,t}$

is called a *star*. For a positive integer  $k$ , let  $[k]$  denote the set  $\{1, 2, \dots, k\}$ . For a set of graphs  $\mathcal{H}$ , let  $\overline{\mathcal{H}} = \{\overline{H} : H \in \mathcal{H}\}$ . For a graph  $G$ , we say  $G$  is *connected* (*anti-connected*) if for every two vertices  $v_1, v_2 \in G$ , there is a path in  $G$  ( $\overline{G}$ , resp.) with ends  $v_1, v_2$ . We say  $S$  is a *connected component* (*anti-connected component*) of  $G$ , if  $S$  is a maximal connected (anti-connected, resp.) induced subgraph of  $G$ . Let  $G$  be a graph, and let  $X, Y \subseteq V(G)$  be disjoint. Let  $E(X, Y)$  be the set of edges with one end in  $X$  and one end in  $Y$ . We say that  $X$  is *complete* (*anti-complete*) to  $Y$ , if every vertex in  $X$  is adjacent (non-adjacent, resp.) to every vertex in  $Y$ . We say that  $X$  is *pure* to  $Y$  if  $X$  is complete or anti-complete to  $Y$ , and  $X$  and  $Y$  are *mixed* otherwise. If  $X = \{x\}$ , we say that  $x$  is mixed on  $Y$ .

For  $c \in (0, 1)$ ,  $X$  is *c-sparse* to  $Y$  if every vertex in  $X$  has at most  $c|Y|$  neighbors in  $Y$ , and  $X$  and  $Y$  are *weakly c-sparse* if  $|E(X, Y)| \leq c|X||Y|$ . For convenience, we say that  $X$  and  $Y$  are *c-sparse* if  $X$  is *c-sparse* to  $Y$  and  $Y$  is *c-sparse* to  $X$ . An  $(\ell, w)$ -*blockade* in a graph  $G$ , is a sequence  $\mathcal{B} = (B_1, \dots, B_k)$  of disjoint subsets of  $V(G)$ , where  $k \geq \ell$  and  $|B_i| \geq w$  for all  $i \in [k]$ . Each  $B_i$  in  $\mathcal{B}$  is called a *block*. Let  $V(\mathcal{B}) = \bigcup_{i=1}^k B_i$ . A blockade  $\mathcal{B} = (B_1, \dots, B_k)$  is *complete* (*anti-complete*, *pure*, *weakly x-sparse*) if its blocks are pairwise complete (anti-complete, pure, weakly  $x$ -sparse, resp.). We say  $\mathcal{B} = (B_1, \dots, B_k)$  is *x-sparse* if  $B_i$  is  $x$ -sparse to  $B_j$  for all  $i, j \in [k]$  with  $i > j$ .

For a positive integer  $\ell$  and  $w \geq 0$ , an  $(\ell, w)$ -*comb* in a graph  $G$ , is a sequence of pairs  $((a_i, B_i), i \in [\ell])$  such that the following holds:

- $(B_1, \dots, B_\ell)$  is an  $(\ell, w)$ -blockade;
- $a_1, \dots, a_\ell$  are distinct;
- $\{a_1, \dots, a_\ell\}, B_1, \dots, B_\ell$  are disjoint subsets of  $V(G)$ ;
- for all distinct  $i, j \in [\ell]$ ,  $a_i$  is complete to  $B_i$  and anti-complete to  $B_j$ .

## 2.1 Leaf-reducibility and wonderfulness

For a finite class of graphs  $\mathcal{F}$ , we say that  $\mathcal{F}$  is *leaf-reducible* if there exists some graph  $H \in \mathcal{F}$  such that  $H$  has a vertex  $v$  of degree 1 and  $(\mathcal{F} \setminus \{H\}) \cup \{(H - v)\}$  satisfies the Erdős-Hajnal property. By Theorem 1.4 and the fact that  $P_4$  has the Erdős-Hajnal property,  $E$ -graph and Bird are leaf-reducible. We say that  $\mathcal{F}$  is *wonderful* if there is a constant  $a \geq 6$  such that the following holds for every  $y \in (0, 1/2)$  and every  $\overline{\mathcal{F}}$ -free graph  $G$ . If  $\mathcal{B} = (B_1, \dots, B_\ell)$  is a  $(\ell, w)$ -blockade where  $\ell$  is an integer at least  $y^{-a}$ , each block of  $\mathcal{B}$  is anti-connected and has the same size, and every two blocks in  $\mathcal{B}$  are complete or  $y^a$ -sparse, then one of the following holds.

- $G$  has a  $y^4$ -restricted induced subgraph of size at least  $w$ .
- There exists  $i \in [\ell]$ , such that there are at most  $y|G|$  vertices  $v \in V(G) \setminus V(\mathcal{B})$  such that  $|N(v) \cap B_i| \in (0, \frac{1}{2}|B_i|)$ .

For a graph  $H$  with two special vertices  $v_1$  and  $v_2$  in  $H$ , let  $H^+$  be the graph obtained from  $H$  by adding a new vertex  $v$ , two edges  $vv_1, vv_2$  and an edge  $v_1v_2$  if  $v_1v_2 \notin E(H)$ , and  $H^-$  be the graph obtained from  $H$  by adding a vertex  $v$  and two edges  $vv_1, vv_2$  and deleting an edge  $v_1v_2$  if  $v_1v_2 \in E(H)$ . We prove that  $E$ -graph and Bird are wonderful.

**Lemma 2.1.** *If  $\mathcal{F}$  satisfies one of the following, then  $\mathcal{F}$  is wonderful.*

- (1)  $\mathcal{F}$  contains an induced subgraph of 1-subdivision of  $K_{1,t}$  for some  $t \geq 1$ .
- (2) There is a graph  $H$  with two distinct special vertices  $v_1, v_2 \in V(H)$  such that  $\{H\} \cup \overline{\mathcal{F}}$  has the Erdős-Hajnal property and neither  $H^+$  nor  $H^-$  is  $\overline{\mathcal{F}}$ -free.

**Proof.** We shall determine  $a$  later. Let  $y \in (0, 1/2)$  and  $G$  be  $\overline{\mathcal{F}}$ -free. Let  $\mathcal{B} = (B_1, \dots, B_\ell)$  be a  $(\ell, w)$ -blockade where  $\ell$  is an integer at least  $y^{-a}$ , each block of  $\mathcal{B}$  is anti-connected and has the same size, and every two blocks are complete or  $y^a$ -sparse.

Let  $I(v)$  be the set of indices  $i \in [\ell]$  such that  $|N(v) \cap B_i| \in (0, \frac{1}{2}|B_i|)$ . If  $|I(v)| \leq y\ell$  for every vertex  $v \in V(G) \setminus V(\mathcal{B})$ , then the second bullet of wonderfulness holds by double counting. So we may suppose that there exists  $v \in V(G) \setminus V(\mathcal{B})$  with  $|I(v)| \geq y\ell$ . Without loss of generality, we may assume that  $I(v) = [|I(v)|]$ . Let  $J$  be the graph with  $V(J) = I(v)$  and  $ij \in E(J)$  if and only if  $B_i$  is complete to  $B_j$ . Since  $|N(v) \cap B_i| \in (0, \frac{1}{2}|B_i|)$  and  $B_i$  is anti-connected for every  $i \in I(v)$ , there exists a non-edge  $b_i b'_i \in B_i$  such that  $v$  is adjacent to  $b_i$  but is non-adjacent to  $b'_i$ .

**Claim 2.1.1.** *Let  $a_0 = 1$  if (1) holds and  $a_0$  be the maximum size of a graph in  $\{H\} \cup \overline{\mathcal{F}}$  if (2) holds. Then there exists a constant  $c \in (0, 1)$  such that if  $a \geq a_0$ , then  $J$  has a clique or stable set of size at least  $|J|^c$ .*

**Proof of Claim 2.1.1.** Suppose first that (1) holds, i.e.,  $\mathcal{F}$  contains an induced subgraph of 1-subdivision of  $K_{1,t}$  for some  $t \geq 1$ . We claim that  $c = t^{-1}$  suffices. If  $J$  has a clique of size  $t$ , say induced by  $[t]$ , then  $\{v, b_1, b'_1, \dots, b_t, b'_t\}$  induces the complement of 1-subdivision of  $K_{1,t}$ . Therefore,  $J$  has no clique of size  $t$  and so has a stable set of size at least  $|J|^{t^{-1}}$  by the Ramsey theorem [11].

Now suppose that (2) holds. Since  $\{H\} \cup \overline{\mathcal{F}}$  has the Erdős-Hajnal property, there is a constant  $c > 0$  such that every  $(\{H\} \cup \overline{\mathcal{F}})$ -free graph of size  $n$  has a clique or a stable set of size at least  $n^c$ . It suffices to prove that  $J$  is  $(\{H\} \cup \overline{\mathcal{F}})$ -free.

Suppose by contradiction that  $J$  has an induced copy of some graph  $F$  in  $\{H\} \cup \overline{\mathcal{F}}$ . For convenience, we may assume that  $V(F) = [s]$ . Suppose first that  $F \in \overline{\mathcal{F}}$ . Let  $v_1$  be a vertex in  $B_1$  and we define  $v_2, \dots, v_s$  inductively as follows. Suppose that  $v_1, \dots, v_{j-1}$  have been defined with  $2 \leq j \leq s$ . Since  $s \leq a_0 \leq a$  and  $a \geq 1$ , we have  $(s-1)y^a \leq (a-1)y^a \leq (a-1)(\frac{1}{2})^a < \frac{1}{2}$ . It follows that  $1 - \frac{1}{2} - (s-1)y^a > 0$ , which implies that one can choose a vertex  $v_j \in B_j$  such that  $v_i v_j \in E(G)$  if and only if  $ij \in E(J)$  for all  $i \in [j-1]$ . So  $\{v_1, \dots, v_s\}$  induces an  $F$  in  $G$ , a contradiction. Now suppose that  $F = H$ . Without loss of generality, we may assume that 1, 2 are the special vertices of  $H$ . Since  $v$  is mixed on  $B_i$  for every  $i \in [s]$ , we may take  $v_1 \in B_1, v_2 \in B_2$  such that  $vv_1, vv_2 \in E(G)$ . We define  $v_3, \dots, v_s$  inductively. Suppose that  $v_1, \dots, v_{j-1}$  are defined where  $3 \leq j \leq s$ . Since  $1 - \frac{1}{2} - (s-1)y^a > 0$ , we could take a vertex  $v_j \in B_j \setminus N(v)$  such that  $v_i v_j \in E(G)$  if and only if  $ij \in E(J)$  for all  $i \in [j-1]$ . Then  $\{v, v_1, \dots, v_s\}$  induces an  $H^+$  or  $H^-$ , depending on whether  $v_1 v_2 \in E(G)$ . This contradicts that neither  $H^+$  nor  $H^-$  is  $\overline{\mathcal{F}}$ -free. ■

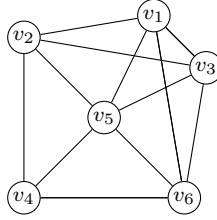
Let  $a = \max\{a_0, \frac{5}{c} + 1\} \geq 6$  where  $a_0, c$  are given by Claim 2.1.1. We claim that  $a$  satisfies Lemma 2.1. Since  $a \geq a_0$ ,  $J$  has a clique or a stable set  $R$  of size  $r = \lceil |J|^c \rceil$  by Claim 2.1.1. By taking complement if necessary, we may assume that  $R$  be a stable set. Let  $S = \bigcup_{i \in R} B_i$ . Since each block of  $\mathcal{B}$  has the same size, each vertex  $v \in B_i \subseteq S$  has at most  $r^{-1}|S|$  neighbors in  $B_i$  and at most  $y^a|S|$  neighbors in  $S \setminus B_i$ . Since

$$\begin{aligned} r^{-1} + y^a &\leq (y\ell)^{-c} + y^a && (r \geq y\ell) \\ &\leq y^{c(a-1)} + y^a && (\ell \geq y^{-a}) \\ &\leq y^5 + y^5 && (a \geq \frac{5}{c} + 1 \geq 6) \\ &\leq y^4, && (y < 1/2) \end{aligned}$$

$S$  is  $y^4$ -sparse. This gives the first outcome of Lemma 2.1.  $\square$

**Lemma 2.2.** *E-graph and Bird are wonderful.*

**Proof.** By Lemma 2.1,  $E$ -graph is wonderful since  $E$ -graph is an induced subgraph of 1-subdivision of  $K_{1,3}$ . Let  $H$  be the graph labeled as in Figure 6. Since  $\{v_1, v_3\}$  is a homogeneous set of  $H$ ,  $H$  has the Erdős–Hajnal property by Theorem 1.4. Let  $H'$  be the graph obtained from  $H$  by adding a new vertex  $v$  and edges  $vv_1$  and  $vv_2$ . It follows that  $H' - v_5$  is a co-bird and  $(H' - v_1v_2) - v_6$  is a co-Bird, which implies that Bird is wonderful by Lemma 2.1.  $\square$



**Fig. 6.** A graph  $H$  with the Erdős–Hajnal property.

## 2.2 Lemmas from literature

We will use several results about restricted induced subgraphs and blockades in a graph. The following lemmas are proved in [15].

**Lemma 2.3.** (Lemma 4.1 of Nguyen-Scott-Seymour [15]) *Let  $k \geq 2$  be an integer. If  $G$  is a graph whose anti-connected components have size less than  $|G|/k$ , then there is a complete  $(k, |G|/k^2)$ -blockade in  $G$ .*

**Lemma 2.4.** (Theorem 7.4 of Nguyen-Scott-Seymour [15]) *Let  $\epsilon \in (0, \frac{1}{2})$  and  $a \geq 1$ . If  $G$  is a graph such that for every induced subgraph  $F$  of  $G$  with  $|F| \geq \epsilon^{2a}|G|$ , there exists a complete or anti-complete  $(k, |F|/k^a)$ -blockade in  $F$  with  $k \in [2, \epsilon^{-1}]$ , then  $G$  has an  $\epsilon$ -restricted induced subgraph with at least  $\epsilon^{3a}|G|$  vertices.*

**Lemma 2.5.** (Theorem 6.1 of Nguyen-Scott-Seymour [15]) *Let  $\epsilon \in (0, \frac{1}{2})$ ,  $d \geq 1$ , and  $x = \epsilon^{5d}$ . If  $G$  is a graph with  $|G| \geq \epsilon^{-10d^2}$  such that for every induced subgraph  $F$  of  $G$  with  $|F| \geq \epsilon^d|G|$ ,*

there exists a pure or  $x$ -sparse  $(k, |F|/k^d)$ -blockade in  $F$  with  $k \in [2, x^{-1}]$ , then there is an  $(\epsilon^{-1}, x^{2d}|G|)$ -blockade  $(B_1, \dots, B_\ell)$  in  $G$  such that for all distinct  $i, j \in [\ell]$ ,  $(B_i, B_j)$  is either complete or weakly  $\epsilon^d$ -sparse in  $G$ .

The following lemmas are implicit in [13, 14, 15]

**Lemma 2.6.** (contained in Lemma 7.1 of Nguyen-Scott-Seymour [15]) *Let  $a \geq 0$ ,  $0 < \epsilon \leq \frac{1}{4}$  and  $m > 0$ . If a graph  $G$  has a blockade  $(A_1, \dots, A_\ell)$  with  $\ell = \lceil \epsilon^{-1} \rceil$  and  $|A_i| \geq m$ , then there is a blockade  $(D_1, \dots, D_\ell)$  such that*

- $D_i$  is a subset of  $A_i$  with  $|D_i| = \lceil \epsilon \lceil m \rceil \rceil$  for every  $i \in [\ell]$ ,
- if  $A_i$  and  $A_j$  are weakly  $\epsilon^a$ -sparse, then  $D_i$  is  $\epsilon^{a-5}$ -sparse to  $D_j$  and  $D_j$  is  $\epsilon^{a-5}$ -sparse to  $D_i$ , and
- if  $A_i$  and  $A_j$  are weakly  $(1 - \epsilon^a)$ -dense, then  $D_i$  is  $(1 - \epsilon^{a-5})$ -dense to  $D_j$  and  $D_j$  is  $(1 - \epsilon^{a-5})$ -dense to  $D_i$ .

**Lemma 2.7.** (contained in Lemma 5.1 of Nguyen-Scott-Seymour [13]) *If a finite set  $\mathcal{F}$  of graphs is leaf-reducible, then there exist constants  $d > 0$  and  $h \geq 1$  such that for every  $y > 0, b > 1$  and every  $y$ -sparse  $\mathcal{F}$ -free graph  $G$ , one of the following holds.*

- There are disjoint  $X, Y \subseteq V(G)$  with  $|X| \geq y^{bd+1}|G|$  and  $|Y| \geq (1 - hy)|G|$  such that  $Y$  is anti-complete to  $X$ .
- $G$  has a  $y^b$ -restricted induced subgraph of size at least  $y^{bd+1}|G|$ .

**Lemma 2.8.** (contained in Lemma 5.3 of Nguyen-Scott-Seymour [15]) *Let  $x \in (0, \frac{1}{2})$ ,  $a > 1, b > 0$ ,  $c = 2^{-4b}$  and  $y \in (0, c]$ . If  $G$  is a graph with  $|G| \geq y^{-(a+2)}$  such that for every induced subgraph  $F$  of  $G$  with  $|F| \geq c|G|$ , there are disjoint  $X, Y \subseteq V(F)$  such that  $|X| \geq y^a|F|$ ,  $|Y| \geq (1 - by)|F|$ , and  $Y$  is  $x$ -sparse or complete to  $X$ , then there is an  $x$ -sparse or complete  $(y^{-1}, y^{a+2}|G|)$ -blockade in  $G$ .*

**Proof of Lemma 2.8.** Let  $n$  be a maximal integer such that  $G$  has a blockade  $(B_1, \dots, B_n)$  such that  $|B_i| \geq y^{a+2}|G|$  for all  $i \in [n]$  and  $|B_n| \geq (1 - by)^n|G|$ , and for all  $i \in [n]$ , either  $B_j$  is  $x$ -sparse to  $B_i$  for all  $j \in [n]$  with  $j > i$  or  $B_j$  is complete to  $B_i$  for all  $j \in [n]$  with  $j > i$ . This is possible since we may take  $n = 1$  and  $B_1 = V(G)$ . Suppose that  $n < 2y^{-1}$ . Note that  $\log(1 - t) \geq -2t \log 2$  for every  $t \in (0, \frac{1}{2})$ . Since  $y \leq c = 2^{-4b}$ ,  $by \leq b2^{-4b} < \frac{1}{2}$ . This implies that  $(1 - by)^{2y^{-1}} = \exp(2y^{-1} \log(1 - by)) \geq \exp(2y^{-1}(-2by \log 2)) = \exp(-4b \log 2) = 2^{-4b} = c$ . Since  $|B_n| \geq (1 - by)^n|G| \geq (1 - by)^{2y^{-1}}|G| \geq 2^{-4b}|G| = c|G|$ , there are disjoint  $X, Y \subseteq B_n$  such that  $|X| \geq y^a|B_n| \geq y^a \cdot c|G| \geq y^{a+2}|G|$ ,  $|Y| \geq (1 - by)|B_n| \geq (1 - by)^{n+1}|G|$ , and  $Y$  is  $x$ -sparse or complete to  $X$ , and for all  $i \in [n - 1]$ ,  $X \cup Y$  is  $x$ -sparse (complete) to  $B_i$  if  $B_n$  is  $x$ -sparse (complete, resp.) to  $B_i$ . So  $(B_1, \dots, B_{n-1}, X, Y)$  contradicts the maximality of  $n$ . This proves that  $n \geq 2y^{-1}$ . Let  $Q, R$  be subsets of  $[n]$  such that  $Q$  is the set of indices  $i$  such that  $B_j$  is  $x$ -sparse to  $B_i$  for all  $j \in [n]$  with  $j > i$  and  $R$  is the set of indices  $i$  such that  $B_j$  is complete to  $B_i$  for all  $j \in [n]$  with  $j > i$ . By the definition of  $(B_1, \dots, B_n)$ , each index  $i \in [n]$  is in one of  $Q, R$ . So one of  $Q, R$  has size at least  $\frac{n}{2} \geq y^{-1}$ . Therefore, one of  $(B_i : i \in Q)$  and  $(B_i : i \in R)$  is a  $x$ -sparse or complete  $(y^{-1}, y^{a+2}|G|)$ -blockade.  $\square$

It is easy to see that Lemma 2.8 still holds if we replace the word “ $x$ -sparse” with “anti-complete”.

**Lemma 2.9.** (contained in Lemma 3.2 of Nguyen-Scott-Seymour [14]) *Let  $c \in (0, 1)$ ,  $b_1 > 1$ ,  $b_2, b_3 > 0$ , and  $b_1 b_2 \geq b_2 + b_3$ . Suppose that  $x \in (0, c)$  and  $G$  is a graph satisfying:*

- *there is a  $c$ -sparse induced subgraph of  $G$  with at least  $c^{b_2}|G|$  vertices, and*
- *for every  $\lambda \in [x, c]$  and every  $\lambda$ -sparse induced subgraph  $F$  of  $G$  with  $|F| \geq \lambda^{b_2}|G|$ , there is a  $\lambda^{b_1}$ -sparse induced subgraph of  $F$  with at least  $\lambda^{b_3}|F|$  vertices.*

*Then  $G$  contains an  $x$ -sparse induced subgraph with at least  $x^{b_1 b_2}|G|$  vertices.*

It is easy to see that Lemma 2.9 still holds if we replace “sparse” throughout by “restricted”.

**Lemma 2.10.** (contained in Lemma 5.2 of Nguyen-Scott-Seymour [15]) *Let  $0 < x \leq y \leq 2^{-8}$ . For every  $y^3$ -sparse graph  $G$  with  $|G| \geq y^{-4}$ , one of the following holds.*

- *There are disjoint  $X, Y \subseteq V(G)$  such that  $|X| \geq y^4|G|$ ,  $|Y| \geq (1 - 4y)|G|$ , and  $Y$  is  $x$ -sparse to  $X$ .*
- *$G$  is  $2y^4$ -sparse.*
- *For some integer  $\ell \in [y^{-1}, x^{-2}]$ , there is an  $(\ell, y^4|G|/\ell^2)$ -comb  $((a_i, B_i) : i \in [\ell])$  in  $G$ , and a vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ .*

### 3 Reducing the Erdős-Hajnal property to the generalized niceness

In this section, we prove Lemma 1.12 which says  $\mathcal{F}$  has the Erdős-Hajnal property if  $\mathcal{F}$  is generalized nice and  $\mathcal{F}$  is leaf-reducible and wonderful. The method used in this section and next section is originated from [15, 14]. Here, we apply and adapt this framework to a setting that goes beyond the cases treated in their work.

A finite set of graphs  $\mathcal{F}$  is *generalized nice* if there exist  $c_1 \geq 3$ ,  $c_2 \geq 8$ ,  $c_3, c_4, c_5, c_8 > 0$ , and  $c_6 \geq 1$ ,  $c_7 \geq 4$  such that for every  $\overline{\mathcal{F}}$ -free graph  $G$  and every  $0 < \epsilon < \frac{1}{2}$ , either

- $G$  has an  $(\epsilon^{-1}, \epsilon^{c_1}|G|)$ -blockade whose blocks are pairwise complete or weakly  $\epsilon^{c_2}$ -sparse;
- $G$  has a clique or stable set of size at least  $(\epsilon^{c_3}|G|)^{c_4}$ ;
- $G$  has a complete or anti-complete  $(k, |G|/k^{c_5})$ -blockade with  $k \geq \epsilon^{-c_6}$ ;
- $G$  has an  $\epsilon^{c_7}$ -restricted induced subgraph  $S$  of size at least  $\epsilon^{c_8}|G|$ .

We now translates the first outcome of generalized niceness to a long complete blockade, a restricted induced subgraph or an anti-complete pair.

**Lemma 3.1.** *Suppose that  $\mathcal{F}$  is generalized nice, leaf-reducible and wonderful. Then there exist constants  $a_1, a_2, a_5 > 0$  and  $a_3 \geq a_4 \geq 4$  such that for every  $0 < y < \frac{1}{2}$  and every  $y$ -restricted  $\overline{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1)  $G$  has a clique or stable set of size at least  $(y^{a_1}|G|)^{a_2}$ .
- (2)  $G$  has a  $y^{a_4}$ -restricted induced subgraph  $S$  with  $|S| \geq y^{a_3}|G|$ .
- (3)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{a_3}})$ -blockade with  $k \geq y^{-1}$ .
- (4) There are disjoint  $X, Y \subseteq V(G)$  such that  $|X| \geq y^{a_3}|G|$ ,  $|Y| \geq (1 - a_5 y)|G|$ , and  $Y$  is anti-complete or complete to  $X$ .

**Proof.** Let  $c_1 \geq 3$ ,  $c_2 \geq 8$ ,  $c_3, c_4, c_5, c_8 > 0$ , and  $c_6 \geq 1$ ,  $c_7 \geq 4$  be given by the generalized niceness of  $\mathcal{F}$ . Let  $d, h > 0$  be given by Lemma 2.7 and  $a \geq 6$  be given by wonderfulness of  $\mathcal{F}$ . We claim that  $a_1 = ac_3$ ,  $a_2 = c_4$ ,  $a_3 = a(c_1 + c_8 + 5) + c_5 + 4d + 1$ ,  $a_4 = 4$  and  $a_5 = 4 + h$  suffice.

If  $|G| \leq y^{-a_3}$ , then the second outcome of Lemma 3.1 holds. So we may assume that  $|G| \geq y^{-a_3}$ . Suppose first that  $\overline{G}$  is  $y$ -sparse. Since  $\mathcal{F}$  is leaf-reducible, it follows from Lemma 2.7 with  $b = 4$  that one of the following holds.

- There are disjoint  $X, Y \subseteq V(G)$  with  $|X| \geq y^{4d+1}|G|$ ,  $|Y| \geq (1 - hy)|G|$ , such that  $Y$  is complete to  $X$ .
- $G$  has a  $y^4$ -restricted induced subgraph of size at least  $y^{4d+1}|G|$ .

They give the second or the fourth outcome of Lemma 3.1. So we may assume that  $G$  is  $y$ -sparse. Let  $\epsilon = y^a$ . By the generalized niceness of  $\mathcal{F}$ , one of the following holds.

- $G$  has an  $(\epsilon^{-1}, \epsilon^{c_1}|G|)$ -blockade whose blocks are pairwise complete or weakly  $\epsilon^{c_2}$ -sparse.
- $G$  has a clique or stable set of size at least  $(\epsilon^{c_3}|G|)^{c_4}$ .
- $G$  has a complete or anti-complete  $(k, |G|/k^{c_5})$ -blockade with  $k \geq \epsilon^{-c_6}$ .
- $G$  has an  $\epsilon^{c_7}$ -restricted induced subgraph  $S$  of size at least  $\epsilon^{c_8}|G|$ .

If the second bullet holds, then the first outcome of Lemma 3.1 holds since  $(\epsilon^{c_3}|G|)^{c_4} \geq (y^{a_1}|G|)^{a_2}$ . If the third bullet holds, then the third outcome of Lemma 3.1 holds since  $\frac{|G|}{k^{c_5}} \geq \frac{|G|}{k^{a_3}}$  and  $\epsilon^{-c_6} \geq y^{-1}$ . If the fourth bullet holds, then the second outcome of Lemma 3.1 holds since  $\epsilon^{c_7} = y^{ac_7} \leq y^{a_4}$  (as  $c_7 \geq 4$ ) and  $\epsilon^{c_8} = y^{ac_8} \geq y^{a_3}$ .

So we may assume that the first bullet holds, that is,  $G$  has a blockade  $(A_1, \dots, A_\ell)$  with  $\ell = \lceil \epsilon^{-1} \rceil$ , and each block has size at least  $m = \epsilon^{c_1}|G| \geq \epsilon^{c_1} \cdot y^{-a_3} \geq \epsilon^{c_1} \cdot \epsilon^{-(c_1+5)} = \epsilon^{-5} > 1$  such that the blocks are pairwise complete or weakly  $\epsilon^{c_2}$ -sparse.

Let  $J$  be the graph with vertices set  $[\ell]$ , and  $ij \in E(J)$  if and only if  $A_i$  is complete to  $A_j$ . Note that  $\epsilon = y^a \leq 2^{-6}$ . By Lemma 2.6, there is a blockade  $(D_1, \dots, D_\ell)$  with  $D_i \subseteq A_i$  such that  $|D_i| = \lceil \epsilon \lceil m \rceil \rceil \geq \epsilon m$  for every  $i \in [\ell]$  and  $D_i$  is  $\epsilon^{c_2-5}$ -sparse to  $D_j$  for every  $ij \in E(\overline{J})$ .

Suppose that for some  $i \in [\ell]$ ,  $D_i$  has no anti-connected components of size at least  $\frac{|D_i|}{\ell}$ . By Lemma 2.3,  $D_i$  contains a complete  $(\ell, \frac{|D_i|}{\ell^2})$ -blockade. Since  $\ell \geq \epsilon^{-1} = y^{-a} \geq y^{-1}$ ,  $\frac{|D_i|}{\ell^2} = \frac{\lceil \epsilon \lceil m \rceil \rceil}{\lceil \epsilon^{-1} \rceil^2} \geq \frac{\epsilon \cdot \epsilon^{c_1} |G|}{\epsilon^{-4}} = \epsilon^{c_1+5} |G| = y^{a(c_1+5)} |G| \geq y^{a_3} |G|$ . This gives the third outcome of Lemma 3.1, a contradiction. So for each  $i \in [\ell]$ ,  $D_i$  has an anti-connected component of size at least  $\frac{|D_i|}{\ell}$ . By anti-connectivity, we may take an anti-connected subset  $B_i \subseteq D_i$  with  $|B_i| = \lceil \frac{|D_i|}{\ell} \rceil$  for all  $i \in [\ell]$ . Note that  $|B_i| \geq \frac{|D_i|}{\ell} \geq \epsilon^2 |D_i| \geq \epsilon^3 m \geq \epsilon^{-2} > 1$ . Since  $D_i$  is  $\epsilon^{c_2-5}$ -sparse to  $D_j$  and  $|B_j| \geq \epsilon^2 |D_j|$ ,  $B_i$  is  $\epsilon^{c_2-7}$ -sparse to  $B_j$  for all  $ij \in E(\bar{J})$ .

Let  $B = V(G) \setminus (\bigcup_{i \in [\ell]} B_i)$ . Since  $\mathcal{F}$  is wonderful and  $\epsilon^{c_2-7} = y^{a(c_2-7)} \leq y^a$  (as  $c_2 \geq 8$ ), one of the following holds.

- $G$  has a  $y^4$ -restricted induced subgraph of size at least  $\epsilon^3 m$ .
- There exists  $i \in [\ell]$ , such that there are at most  $y|G|$  vertices  $v \in V(G) \setminus V(\mathcal{B})$  with  $|N(v) \cap B_i| \in (0, \frac{1}{2}|B_i|)$ .

If  $G$  has a  $y^4$ -restricted induced subgraph of size at least  $\epsilon^3 m \geq y^{a(c_1+3)} |G|$ , then the second outcome of Lemma 3.1 holds. So we may assume that there exists  $i \in [\ell]$  such that there are at most  $y|G|$  vertices  $v \in V(G) \setminus V(\mathcal{B})$  such that  $|N(v) \cap B_i| \in (0, \frac{1}{2}|B_i|)$ .

Since  $G$  is  $y$ -sparse, there are at most  $2y|G|$  vertices in  $B$  that has at least  $\frac{|B_i|}{2}$  neighbors in  $B_i$ . Since  $m > 1$ ,  $|D_i| \leq \lceil m \rceil \leq 2m$ . Since  $\frac{|D_i|}{\ell} > 1$ ,  $|B_i| \leq 2 \frac{|D_i|}{\ell}$ . This implies that

$$\begin{aligned} \sum_{i=1}^{\ell} |B_i| &\leq \ell \cdot \frac{4m}{\ell} = 4m \\ &\leq \epsilon^{-2} \cdot \epsilon^{c_1} |G| && (0 < \epsilon < 1/2) \\ &\leq \epsilon |G| && (c_1 \geq 3) \\ &\leq y |G|. && (\epsilon \leq y) \end{aligned}$$

Let  $Y$  be the set of vertices in  $D$  that is anti-complete to  $D_i$ . Then  $|Y| \geq |G| - \sum_{i=1}^{\ell} |B_i| - y|G| - 2y|G| \geq (1 - 4y)|G|$ . Since  $|B_i| \geq \epsilon^3 m = \epsilon^{c_1+3} |G| = y^{a(c_1+3)} |G| \geq y^{a_3} |G|$ ,  $B_i$  and  $Y$  satisfy the fourth outcome of Lemma 3.1.

This complete the proof of Lemma 3.1.  $\square$

We then present a lemma to translate the fourth outcome of Lemma 3.1 to an anti-complete or complete blockade.

**Lemma 3.2.** *Suppose that  $\mathcal{F}$  is generalized nice, leaf-reducible and wonderful. Then there exist constants  $c \in (0, \frac{1}{2})$ ,  $c_1, c_2 > 0$  and  $c_3 \geq c_4 \geq 4$  such that for  $y \in (0, c]$  and every  $cy$ -restricted  $\bar{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1)  $G$  has a clique or stable set of size at least  $(y^{c_1} |G|)^{c_2}$ .
- (2)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{c_3}})$ -blockade with  $k \geq y^{-1}$ .
- (3)  $G$  has a  $y^{c_4}$ -restricted induced subgraph  $S$  with  $|S| \geq y^{c_3} |G|$ .

**Proof.** Let  $a_1, a_2, a_5 > 0$  and  $a_3 \geq a_4 \geq 4$  be given by Lemma 3.1. We claim that  $c = 2^{-4a_5}$ ,  $c_1 = a_1 + 1$ ,  $c_2 = a_2$ ,  $c_3 = a_3 + 2$  and  $c_4 = a_4$  suffice. If  $|G| \leq y^{-c_3}$ , then the third outcome of Lemma 3.2 holds. So we may assume that  $|G| \geq y^{-c_3}$ .

Suppose first that for every induced subgraph  $F$  of  $G$  with  $|F| \geq c|G|$ , there are disjoint  $X, Y \subseteq V(F)$  such that  $|X| \geq y^{a_3}|F|$ ,  $|Y| \geq (1 - a_5y)|F|$ , and  $Y$  is anti-complete or complete to  $X$ . By Lemma 2.8 with  $a = a_3$ ,  $b = a_5$ , and “ $x$ -sparse” replaced by “anti-complete”, there is an anti-complete or complete  $(y^{-1}, y^{a_3+2}|G|)$ -blockade in  $G$ , which gives the second outcome of Lemma 3.2.

Therefore, we may assume that there is an induced subgraph  $F$  of  $G$  with  $|F| \geq c|G|$  such that there is no disjoint  $X, Y \subseteq V(F)$  such that  $|X| \geq y^{a_3}|F|$ ,  $|Y| \geq (1 - a_5y)|F|$ , and  $Y$  is anti-complete or complete to  $X$ .

Since  $G$  is  $cy$ -restricted,  $F$  is  $y$ -restricted. By Lemma 3.1 with  $G$  replaced by  $F$ , one of the following holds.

- $F$  has a clique or stable set of size at least  $(y^{a_1}|F|)^{a_2}$ .
- $F$  has a  $y^{a_4}$ -restricted induced subgraph  $S$  with  $|S| \geq y^{a_3}|F|$ .
- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{a_3}})$ -blockade with  $k \geq y^{-1}$ .

If the first bullet holds, then the first outcome of Lemma 3.2 holds since  $(y^{a_1}|F|)^{a_2} \geq (y^{a_1+1}|G|)^{a_2} = (y^{c_1}|G|)^{c_2}$ . If the second bullet holds, then the third outcome of Lemma 3.2 holds since  $y^{a_3}|F| \geq y^{a_3+1}|G| \geq y^{c_3}|G|$  and  $a_4 = c_4$ . If the third bullet holds, then the second outcome of Lemma 3.2 holds since  $\frac{|F|}{k^{a_3}} \geq \frac{|G|}{y^{-1} \cdot k^{a_3}} \geq \frac{|G|}{k^{a_3+1}} \geq \frac{|G|}{k^{c_3}}$ .

This complete the proof of Lemma 3.2 □

We then use the method of iterative sparsification to decrease the sparse condition of Lemma 3.2 to a constant sparse condition.

**Lemma 3.3.** *Suppose that  $\mathcal{F}$  is generalized nice, leaf-reducible and wonderful. Then there exist constants  $c \in (0, \frac{1}{2})$ ,  $a_1 \geq 1$ ,  $a_2 > 0$  such that for every  $x \in (0, c^2)$  and every  $c^2$ -restricted  $\overline{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1)  $G$  has an  $x$ -restricted induced subgraph  $S$  with  $|S| \geq x^{a_1}|G|$ .
- (2)  $G$  has a clique or stable set of size at least  $(x^{a_1}|G|)^{a_2}$ .
- (3)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{a_1}})$ -blockade with  $k \geq 2$ .

**Proof.** Let  $c \in (0, \frac{1}{2})$ ,  $c_1, c_2 > 0$  and  $c_3 \geq c_4 \geq 4$  be given by Lemma 3.2. We claim that  $c, a_1 = c_1 + 3c_3$  and  $a_2 = c_2$  suffice. Suppose that none of outcomes of Lemma 3.3 holds. We first present a claim to give the condition of Lemma 2.9.

**Claim 3.3.1.** *For every  $y$  with  $cy \in [x, c^2]$  and every  $cy$ -restricted induced subgraph  $F$  of  $G$  with  $|F| \geq (cy)^{\frac{4c_3}{c_4}}|G|$ , there is a  $(cy)^{\frac{c_4}{2}}$ -restricted induced subgraph of  $F$  with at least  $(cy)^{c_3}|F|$  vertices.*

**Proof of Claim 3.3.1.** By Lemma 3.2 with  $G$  replaced by  $F$ , one of the following holds.

- $F$  has a clique or stable set of size at least  $(y^{c_1}|F|)^{c_2}$ .
- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{c_3}})$ -blockade with  $k \geq y^{-1}$ .
- $F$  has a  $y^{c_4}$ -sparse induced subgraph  $S$  with  $|S| \geq y^{c_3}|F|$ .

If the first bullet holds, then the second outcome of Lemma 3.3 holds since  $(y^{c_1}|F|)^{c_2} \geq (y^{c_1} \cdot (cy)^{\frac{4c_3}{c_4}}|G|)^{c_2} \geq (y^{c_1} \cdot (y^2)^{\frac{4c_3}{c_4}}|G|)^{c_2} = (y^{c_1 + \frac{8c_3}{c_4}}|G|)^{c_2} \geq (x^{c_1 + \frac{8c_3}{c_4}}|G|)^{c_2} \geq (x^{a_1}|G|)^{a_2}$ , where the last inequality is due to  $c_4 \geq 4$ .

If the second bullets holds, then the third outcome of Lemma 3.3 holds since  $\frac{|F|}{k^{c_3}} \geq \frac{y^{\frac{8c_3}{c_4}}|G|}{k^{c_3}} \geq \frac{|G|}{k^{c_3 + \frac{8c_3}{c_4}}} \geq \frac{|G|}{k^{\frac{c_4}{c_3} + \frac{8c_3}{c_4}}} \geq \frac{|G|}{k^{a_1}}$ .

So we may assume that the third outcome holds. Since  $y^{c_4} = (y^2)^{\frac{c_4}{2}} \leq (cy)^{\frac{c_4}{2}}$  and  $y^{c_3} \geq (cy)^{c_3}$ , Claim 3.3.1 follows. ■

Let  $b_1 = \frac{c_4}{2}, b_2 = \frac{4c_3}{c_4}, b_3 = c_3$ . Since  $c_4 \geq 4$ ,  $b_1b_2 = 2c_3 \geq c_3 + \frac{4c_3}{c_4} = b_2 + b_3$ . By Claim 3.3.1 and Lemma 2.9 (with  $b_1, b_2, b_3$  and  $\lambda$  replaced by  $cy$ ,  $c$  replaced by  $c^2$ , sparse replaced by restricted),  $G$  has an  $x$ -restricted induced subgraph with at least  $x^{2c_3}|G|$  vertices. Since  $x^{2c_3} \geq x^{a_1}$ , the first outcome of Lemma 3.3 holds.

This completes the proof of Lemma 3.3. □

We then present a lemma to obtain the polynomial Rödl property.

**Lemma 3.4.** *Suppose that  $\mathcal{F}$  is generalized nice, leaf-reducible and wonderful. Then there exist constants  $c_1 \geq 1, c_2 > 0$  such that for every  $x \in (0, \frac{1}{2})$  and for every  $\overline{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1)  $G$  has an  $x$ -restricted induced subgraph  $S$  with  $|S| \geq x^{c_1}|G|$ .
- (2)  $G$  has a complete or anti-complete  $(k, |G|/k^{c_1})$ -blockade with  $k \geq 2$ .
- (3)  $G$  has a clique or stable set of size at least  $(x^{c_1}|G|)^{c_2}$ .

**Proof.** Let  $c \in (0, \frac{1}{2}), a_1 \geq 1$  and  $a_2 > 0$  be given by Lemma 3.3. Let  $\xi = c^2$ . By Theorem 1.1 (with  $\epsilon = \xi$ ), there exists  $\delta > 0$  such that every  $\overline{\mathcal{F}}$ -free graph  $G$  has a  $\xi$ -restricted induced subgraph of size at least  $\delta|G|$ . Let  $c_2 = a_2$  and  $c_1$  be sufficiently large such that  $\delta \geq \xi^{\frac{c_1}{2}}$ ,  $\delta \geq 2^{a_1 - c_1}$  and  $c_1 \geq 2a_1$ . In the following we prove that  $c_1, c_2$  suffice.

By Theorem 1.1,  $G$  has a  $\xi$ -restricted induced subgraph  $F$  with  $|F| \geq \delta|G|$ . If  $x \geq \xi$ , then  $F$  is  $x$ -restricted with  $|F| \geq \delta|G| \geq 2^{a_1 - c_1}|G| \geq (x^{-1})^{a_1 - c_1}|G| = x^{c_1 - a_1}|G| \geq x^{c_1}|G|$ , which gives the first outcome of Lemma 3.4. So we may assume that  $x < \xi = c^2$ . By Lemma 3.3 with  $G$  replaced by  $F$ , one of the following holds.

- $F$  has an  $x$ -restricted induced subgraph  $S$  with  $|S| \geq x^{a_1}|F|$ .
- $F$  has a clique or stable set of size at least  $(x^{a_1}|F|)^{a_2}$ .

- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{a_1}})$ -blockade with  $k \geq 2$ .

Note that

$$\begin{aligned}
x^{a_1}|F| &\geq x^{a_1}\delta|G| & (|F| \geq \delta|G|) \\
&\geq x^{a_1}\xi^{\frac{c_1}{2}}|G| & (\delta \geq \xi^{\frac{c_1}{2}}) \\
&\geq x^{a_1+\frac{c_1}{2}}|G| & (\xi > x) \\
&\geq x^{c_1}|G|. & (2a_1 \leq c_1)
\end{aligned}$$

If the first bullet holds, then since  $|S| \geq x^{a_1}|F| \geq x^{c_1}|G|$ , the first outcome of Lemma 3.4 holds. If the second bullet holds, then since  $G$  has a clique or stable set of size at least  $(x^{a_1}|F|)^{a_2} \geq (x^{c_1}|G|)^{c_2}$ , the third outcome of Lemma 3.4 holds. So we may assume that the third bullet holds. Since  $\delta \geq 2^{a_1-c_1}$  and  $k \geq 2$ , we have  $\frac{|F|}{k^{a_1}} \geq \frac{\delta|G|}{k^{a_1}} \geq \frac{2^{a_1-c_1}|G|}{k^{a_1}} \geq \frac{|G|}{k^{c_1}}$ . Therefore, the second outcome of Lemma 3.4 holds. This complete the proof of Lemma 3.4.  $\square$

Now we are ready to prove Lemma 1.12, which we restate here for readers' convenience.

**Lemma 3.5.** *Let  $\mathcal{F}$  be a finite class of graphs that is leaf-reducible and wonderful. If  $\mathcal{F}$  is generalized nice, then  $\mathcal{F}$  has the Erdős-Hajnal property.*

**Proof.** Let  $c_1 \geq 1, c_2 > 0$  be given by Lemma 3.4. Let  $q = 42c_1^2$  and  $m = 2^q$ ,  $c = \min\{q^{-1}, \frac{c_2}{2}\}$ . We claim that every  $\overline{\mathcal{F}}$ -free graph  $G$  has a clique or stable set of size at least  $|G|^c$ . Suppose to the contrary that  $G$  has no clique or stable set of size at least  $|G|^c$ . Since every graph on two or more vertices has a clique or stable set of size 2, we may assume that  $|G| > m$ .

Let

$$x = |G|^{-\frac{1}{3c_1}} \leq m^{-\frac{1}{3c_1}} = 2^{-\frac{42c_1^2}{3c_1}} = 2^{-14c_1}$$

and

$$\epsilon = x^{\frac{1}{7c_1}} \leq 2^{-\frac{14c_1}{7c_1}} = \frac{1}{4}.$$

Since  $c_1 \geq 1$ , we have  $x < \epsilon \leq \frac{1}{4}$ . We then present a claim to give some property of an induced subgraph of  $G$  with at least  $\epsilon^{2c_1}|G|$  vertices.

**Claim 3.5.1.** *Every induced subgraph  $F$  of  $G$  with  $|F| \geq \epsilon^{2c_1}|G|$  has a complete or anti-complete  $(k, |F|/k^{c_1})$ -blockade with  $k \in [2, \epsilon^{-1}]$ .*

**Proof of Claim 3.5.1.** Suppose to the contrary that there is an induced subgraph  $F$  of  $G$  with  $|F| \geq \epsilon^{2c_1}|G|$  such that  $F$  has no complete or anti-complete  $(k, |F|/k^{c_1})$ -blockades with  $k \in [2, \epsilon^{-1}]$ . By Lemma 3.4 with  $G$  replaced by  $F$ , one of the following holds.

- $G$  has an  $x$ -restricted induced subgraph  $S$  with  $|S| \geq x^{c_1}|F|$ .
- $G$  has a complete or anti-complete  $(k, |F|/k^{c_1})$ -blockade with  $k \geq \epsilon^{-1}$ .
- $G$  has a clique or stable set of size at least  $(x^{c_1}|F|)^{c_2}$ .

Suppose first that the first bullet holds. Since  $x = |G|^{-\frac{1}{3c_1}}$  and  $a_1 \geq 1$ ,  $|S| \geq x^{c_1}|F| \geq x^{c_1+\frac{2}{7}}|G| = x^{c_1+\frac{2}{7}} \cdot x^{-3c_1} \geq x^{-1}$ . By taking complement if necessary, we may assume that  $S$  is  $x$ -sparse. Since  $x < \frac{1}{4}$ ,  $S$  has a stable set of size at least  $\frac{|S|}{x|S|+1} = \frac{1}{x+\frac{1}{|S|}} \geq \frac{1}{x+x} \geq x^{-\frac{1}{2}} = |G|^{\frac{1}{6c_1}} \geq |G|^c$ .

We then suppose that the second bullet holds. By taking a vertex from each block, we have a clique or stable set of size  $k \geq \epsilon^{-1} = |G|^{\frac{1}{21c_1^2}} \geq |G|^c$ .

Finally, we suppose that the third bullet holds. So  $G$  has a clique or stable set of size at least  $(x^{c_1}|F|)^{c_2} \geq (x^{c_1}\epsilon^{2c_1}|G|)^{c_2} = (x^{c_1+\frac{2}{7}}|G|)^{c_2} = |G|^{c_2(1-\frac{c_1+\frac{2}{7}}{3c_1})} \geq |G|^{c_2/2} \geq |G|^c$ , where the second last inequality is due to  $c_1 \geq 1$ .

This proves Claim 3.5.1. ■

By Claim 3.5.1 together with Lemma 2.4 (with  $a = c_1$ ),  $G$  has an  $\epsilon$ -restricted induced subgraph  $S$  with  $|S| \geq \epsilon^{3c_1}|G|$ . Since  $\epsilon = x^{\frac{1}{7c_1}} = |G|^{-\frac{1}{21c_1^2}}$  and  $c_1 \geq 1$ ,  $|S| \geq \epsilon^{3c_1}|G| = \epsilon^{-21c_1^2+3c_1} \geq \epsilon^{-1}$ . By taking complement if necessary, we may assume that  $S$  is  $\epsilon$ -sparse. Since  $\epsilon \leq \frac{1}{4}$ ,  $S$  has a stable set of size at least  $\frac{|S|}{\epsilon|S|+1} = \frac{1}{\epsilon+\frac{1}{|S|}} \geq \frac{1}{\epsilon+\epsilon} \geq \epsilon^{-\frac{1}{2}} = |G|^{\frac{1}{42c_1^2}} \geq |G|^c$ .

This completes the proof of Lemma 3.5. □

## 4 Reducing generalized niceness to property (\*)

In this section, we prove Lemma 1.13 which says  $\mathcal{F}$  is generalized nice if  $\mathcal{F}$  has property (\*) and  $\mathcal{F}$  is leaf-reducible.

A finite class of graphs  $\mathcal{F}$ , we say  $\mathcal{F}$  has *property (\*)* if there exist  $c_1, c_2, c_3 > 0$  such that for every  $\overline{\mathcal{F}}$ -free graph  $G$  the following holds. If there is a  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$ , where  $\ell, w \geq 4$ , and there exists  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ , then one of the following holds.

- $G$  has a clique or stable set of size at least  $w^{c_1}$ .
- $G$  has a complete or anti-complete  $(k, w/k^{c_2})$ -blockade with  $k \geq \ell^{c_3}$ .
- $G$  has a pure  $(\ell, w/\ell^2)$ -blockade.

We now prove the first lemma through the property (\*), while the existence of the comb and the special vertex in the statement of property (\*) is given by Lemma 2.10.

**Lemma 4.1.** *Suppose that  $\mathcal{F}$  has property (\*) and  $\mathcal{F}$  is leaf-reducible. Then there exist constants  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 \geq 4$  such that for every  $0 < x \leq y \leq 2^{-4c_5}$  and every  $y^3$ -restricted  $\overline{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1) *There are disjoint  $X, Y \subseteq V(G)$  such that  $|X| \geq y^{c_4}|G|$ ,  $|Y| \geq (1 - c_5y)|G|$ , and  $Y$  is  $x$ -sparse or complete to  $X$ .*

- (2)  $G$  has a  $2y^4$ -restricted induced subgraph of size at least  $y^{c_4}|G|$ .
- (3)  $G$  has a clique or stable set of size at least  $(x^9|G|)^{c_1}$ .
- (4)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{c_2 + \frac{6}{c_3}}})$ -blockade with  $k \geq y^{-c_3}$ .
- (5)  $G$  has a pure  $(\ell, |G|/\ell^8)$  blockade with  $\ell \in [y^{-1}, x^{-2}]$ .

**Proof.** Let  $c_1, c_2, c_3 > 0$  be given by the property (\*). Let  $c_4 = \max\{4d + 1, 4\}$  and  $c_5 = \max\{h, 4\}$ , where  $d, h$  are given by Lemma 2.7. We claim that  $c_1, c_2, c_3, c_4, c_5$  suffice.

Suppose first that  $\overline{G}$  is  $y^3$ -sparse. By Lemma 2.7 with  $b = 4$ , we have

- there are disjoint  $X, Y \subseteq V(G)$  with  $|X| \geq y^{4d+1}$ ,  $|Y| \geq (1 - hy)|G|$ , such that  $Y$  is complete to  $X$  in  $G$  or
- $G$  has a  $y^4$ -restricted induced subgraph of size at least  $y^{4d+1}|G|$ ,

which gives the first or the second outcome of Lemma 4.1.

So we may assume that  $G$  is  $y^3$ -sparse. If  $|G| \leq x^{-9}$ , then the third outcome of Lemma 4.1 holds since  $c_1 > 0$ . So we may assume that  $|G| \geq x^{-9} \geq y^{-4}$ . By Lemma 2.10, one of the following holds.

- There are disjoint  $X, Y \subseteq V(G)$  such that  $|X| \geq y^4|G|$ ,  $|Y| \geq (1 - 4y)|G|$ , and  $Y$  is  $x$ -sparse to  $X$ .
- $G$  is  $2y^4$ -sparse.
- For some integer  $\ell \in [y^{-1}, x^{-2}]$ , there is an  $(\ell, y^4|G|/\ell^2)$ -comb  $((a_i, B_i) : i \in [\ell])$  in  $G$ , and a vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ .

If the first or the second outcome of Lemma 2.10 holds, then the first or the second outcome of Lemma 4.1 holds. So we may assume that the third outcome of Lemma 2.10 holds. Since  $y \leq 2^{-4c_5} \leq 2^{-8}$ ,  $\ell \geq y^{-1} > 4$ . Moreover,

$$\begin{aligned}
 w := y^4|G|/\ell^2 &\geq \max \left\{ \frac{|G|}{\ell^6}, x^8|G| \right\} && (y^{-1} \leq \ell, \ell \leq x^{-2}, y \geq x) \\
 &\geq x^{-1} && (|G| \geq x^{-9}) \\
 &\geq 4 && (x \leq 2^{-4c_5}, c_5 \geq 4)
 \end{aligned}$$

By the condition that  $\mathcal{F}$  has property (\*), one of the following holds.

- $G$  has a clique or stable set of size at least  $w^{c_1}$ .
- $G$  has a complete or anti-complete  $(k, \frac{w}{k^{c_2}})$ -blockade with  $k \geq \ell^{c_3}$ .
- $G$  has a pure  $(\ell, \frac{w}{\ell^2})$ -blockade.

If the first bullet holds, then since  $w^{c_1} \geq (x^8|G|)^{c_1}$ , the third outcome of Lemma 4.1 holds. If the second bullet holds, then since  $\frac{w}{k^{c_2}} \geq \frac{|G|}{\ell^6 \cdot k^{c_2}} \geq \frac{|G|}{k^{c_2 + \frac{6}{c_3}}}$  and  $k \geq \ell^{c_3} \geq y^{-c_3}$ , the fourth outcome of Lemma 4.1 holds. If the third bullet holds, then since  $\frac{w}{\ell^2} \geq \frac{|G|}{\ell^8}$ , the fifth outcome of Lemma 4.1 holds.

This completes the proof of Lemma 4.1.  $\square$

We then present a lemma to translate the first outcome of Lemma 4.1 to a long sparse or complete blockade.

**Lemma 4.2.** *Suppose that  $\mathcal{F}$  has property  $(*)$  and  $\mathcal{F}$  is leaf-reducible. Then there exist constants  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 > 4$  such that for every  $0 < x \leq y \leq c = 2^{-4c_5}$  and every  $cy^3$ -restricted  $\overline{\mathcal{F}}$ -free graph  $G$ , one of the following holds.*

- (1)  $G$  has an  $x$ -sparse or complete  $(y^{-1}, y^{c_4+2}|G|)$ -blockade.
- (2)  $G$  has a  $2y^4$ -restricted induced subgraph  $S$  with  $|S| \geq y^{c_4+2}|G|$ .
- (3)  $G$  has a clique or stable set of size at least  $(x^{10}|G|)^{c_1}$ .
- (4)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{c_2 + \frac{6}{c_3}}})$ -blockade with  $k \geq y^{-c_3}$ .
- (5)  $G$  has a pure  $(\ell, \frac{|G|}{\ell^9})$  blockade with  $\ell \in [y^{-1}, x^{-2}]$ .

**Proof.** Let  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 \geq 4$  be given by Lemma 4.1. We claim that  $c_1, c_2, c_3, c_4, c_5$  suffice. If  $|G| \leq y^{-(c_4+2)}$ , then the second outcome of Lemma 4.2 holds. So we may assume that  $|G| \geq y^{-(c_4+2)}$ .

Suppose first that for every induced subgraph  $F$  of  $G$  with  $|F| \geq c|G|$ , there are disjoint  $X, Y \subseteq V(F)$  such that  $|X| \geq y^{c_4}|F|$ ,  $|Y| \geq (1 - c_5y)|F|$ , and  $Y$  is  $x$ -sparse or complete to  $X$ . By Lemma 2.8 (with  $a = c_4$ ,  $b = c_5$ ), there is an  $x$ -sparse or complete  $(y^{-1}, y^{c_4+2}|G|)$ -blockade. This give the first outcome of Lemma 4.2.

So we may assume that there is an induced subgraph  $F$  of  $G$  with  $|F| \geq c|G|$  such that there is no disjoint  $X, Y \subseteq V(F)$  such that  $|X| \geq y^{c_4}|F|$ ,  $|Y| \geq (1 - c_5y)|F|$ , and  $Y$  is  $x$ -sparse or complete to  $X$ . Since  $G$  is  $cy^3$ -restricted and  $|F| \geq c|G|$ ,  $F$  is  $y^3$ -restricted. By Lemma 4.1 with  $G$  replaced by  $F$ , one of the following holds.

- $F$  has a  $2y^4$ -restricted induced subgraph of size at least  $y^{c_4}|F|$ .
- $F$  has a clique or stable set of size at least  $(x^9|F|)^{c_1}$ .
- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{c_2 + \frac{6}{c_3}}})$ -blockade with  $k \geq y^{-c_3}$ .
- $F$  has a pure  $(\ell, |F|/\ell^8)$  blockade with  $\ell \in [y^{-1}, x^{-2}]$ .

If the first bullet holds, then the second outcome of Lemma 4.2 holds since  $y^{c_4}|F| \geq y^{c_4} \cdot y|G| = y^{c_4+1}|G|$ . If the second bullet holds, then the third outcome of Lemma 4.2 holds since  $(x^9|F|)^{c_1} \geq (x^9 \cdot x|G|)^{c_1} = (x^{10}|G|)^{c_1}$ . If the third bullet holds, then the fourth outcome of

Lemma 4.2 holds since  $\frac{|F|}{k^{c_2+\frac{6}{c_3}}} \geq \frac{y|G|}{k^{c_2+\frac{6}{c_3}}} \geq \frac{|G|}{k^{c_2+\frac{6}{c_3}+\frac{1}{c_3}}} = \frac{|G|}{k^{c_2+\frac{7}{c_3}}}$ . If the fourth bullet holds, then the fifth outcome of Lemma 4.2 holds since  $\frac{|F|}{\ell^8} \geq \frac{y|G|}{\ell^8} \geq \frac{|G|}{\ell^9}$ .

This completes the proof of Lemma 4.2.  $\square$

We then use the method of iterative sparsification to decrease the sparse condition of Lemma 4.2 to a constant sparse condition.

**Lemma 4.3.** *Suppose that  $\mathcal{F}$  has property  $(*)$  and  $\mathcal{F}$  is leaf-reducible. Then there exist constants  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 \geq 4$  such that for every  $0 < x \leq c^{10}$  and every  $c^{10}$ -restricted  $\overline{\mathcal{F}}$ -free graph  $G$  where  $c = 2^{-4c_5}$ , one of the following holds.*

- (1)  $G$  has an  $x$ -restricted induced subgraph  $S$  of size at least  $x^{22c_4}|G|$ .
- (2)  $G$  has a clique or stable set of size at least  $(x^{30c_4}|G|)^{c_1}$ .
- (3)  $G$  has a complete or anti-complete  $(k, \frac{|G|}{k^{c_2+\frac{27c_4}{c_3}}})$ -blockade with  $k \geq 2$ .
- (4)  $G$  has an  $x$ -sparse or pure  $(\ell, |G|/\ell^{29c_4})$  blockade with  $\ell \in [c^{-1}, x^{-2}]$ .

**Proof.** Let  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 \geq 4$  be given by Lemma 4.2. We claim that  $c_1, c_2, c_3, c_4, c_5$  suffice. Suppose that none of Lemma 4.3 holds. We first present a claim to give the condition of Lemma 2.9.

**Claim 4.3.1.** *For every  $y \in [x, c^3]$  and every  $y^{\frac{10}{3}}$ -restricted induced subgraph  $F$  of  $G$  with  $|F| \geq y^{10(c_4+2)}|G|$ , there is a  $y^{\frac{11}{3}}$ -restricted induced subgraph of  $F$  with at least  $y^{(c_4+2)}|F|$  vertices.*

**Proof of Claim 4.3.1.** Suppose to the contrary that there is an  $y^{\frac{10}{3}}$ -restricted induced subgraph  $F$  of  $G$  with  $|F| \geq y^{10(c_4+2)}|G|$  such that  $F$  has no  $y^{\frac{11}{3}}$ -restricted induced subgraph of  $F$  with at least  $y^{(c_4+2)}|F|$  vertices. Since  $y \leq c^3$ ,  $y^{\frac{10}{3}} \leq cy^3$ . By Lemma 4.2 with  $G$  replaced by  $F$ , one of the following holds.

- $F$  has an  $x$ -sparse or complete  $(y^{-1}, y^{c_4+2}|F|)$ -blockade.
- $F$  has a  $2y^4$ -restricted induced subgraph  $S$  with  $|S| \geq y^{c_4+2}|F|$ .
- $F$  has a clique or stable set of size at least  $(x^{10}|F|)^{c_1}$ .
- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{c_2+\frac{7}{c_3}}})$ -blockade with  $k \geq y^{-c_3}$ .
- $F$  has a pure  $(\ell, \frac{|F|}{\ell^9})$  blockade with  $\ell \in [y^{-1}, x^{-2}]$ .

If the first bullet holds, then the fourth outcome of Lemma 4.3 holds since  $y^{-1} \in [c^{-1}, x^{-1}]$  and  $y^{c_4+2}|F| \geq y^{c_4+2} \cdot y^{10(c_4+2)}|G| \geq y^{29c_4}|G|$ .

If the second bullet holds, then it contradicts the condition of Claim 4.3.1 since  $2y^4 \leq y^{\frac{11}{3}}$ .

If the third bullet holds, then the second outcome of Lemma 4.3 holds since  $x^{10}|F| \geq x^{10} \cdot y^{10(c_4+2)}|G| \geq x^{30c_4}|G|$ .

Suppose the fourth bullet holds. If  $k \geq 2$ , then the third outcome of Lemma 4.3 holds since  $\frac{|F|}{k^{c_2 + \frac{7}{c_3}}} \geq \frac{y^{10(c_4+2)}|G|}{k^{c_2 + \frac{7}{c_3}}} \geq \frac{|G|}{k^{c_2 + \frac{7}{c_3} + \frac{10(c_4+2)}{c_3}}} \geq \frac{|G|}{k^{c_2 + \frac{27c_4}{c_3}}}$ . If  $2 > k \geq y^{-c_3} > 1$ , then since  $\frac{|G|}{k^{c_2 + \frac{27c_4}{c_3}}} \geq \frac{|G|}{2^{c_2 + \frac{27c_4}{c_3}}}$ , there is a complete or anti-complete  $(2, \frac{|G|}{2^{c_2 + \frac{27c_4}{c_3}}})$ -blockade, which also gives the third outcome of Lemma 4.3.

If the fifth bullet holds, then the fourth outcome of Lemma 4.3 holds since  $\frac{|F|}{\ell^9} \geq \frac{y^{10(c_4+2)}|G|}{\ell^9} \geq \frac{|G|}{\ell^{29c_4}}$ .

This completes the proof of Claim 4.3.1.  $\blacksquare$

By Claim 4.3.1 and Lemma 2.9 (with  $b_1 = \frac{11}{10}$ ,  $b_2 = 3(c_4 + 2)$ ,  $b_3 = \frac{3(c_4+2)}{10}$  and  $c$  replaced by  $c^{10}$ ,  $x$  replaced by  $x^{\frac{10}{3}}$ ,  $\lambda$  replaced by  $y^{\frac{10}{3}}$ , sparse replaced restricted),  $G$  has an  $x^{\frac{10}{3}}$ -restricted induced subgraph  $S$  with  $|S| \geq (x^{\frac{10}{3}})^{\frac{33(c_4+2)}{10}}|G| = x^{11(c_4+2)}|G| \geq x^{22c_4}|G|$ , which gives the first outcome of Lemma 4.3.

This completes the proof of Lemma 4.3.  $\square$

We then remove the sparsity hypothesis in Lemma 4.3 by using Rödl's Theorem.

**Lemma 4.4.** *Suppose that  $\mathcal{F}$  has property  $(*)$  and  $\mathcal{F}$  is leaf-reducible. Then there exist constants  $c_1 > 0, c_4 \geq 4$  and  $d \geq 58c_4$  such that for every  $x \in (0, 2^{-d})$  and every  $\overline{\mathcal{F}}$ -free graph  $G$  with  $|G| \geq x^{-d}$ , one of the following holds.*

- (1)  $G$  has an  $x$ -restricted induced subgraph  $S$  of size at least  $x^{23c_4}|G|$ .
- (2)  $G$  has a pure or  $x$ -sparse  $(k, |G|/k^d)$ -blockade with  $k \in [2, x^{-1}]$ .
- (3)  $G$  has a clique or stable set of size at least  $(x^{31c_4}|G|)^{c_1}$ .
- (4)  $G$  has a complete or anti-complete  $(k, |G|/k^d)$ -blockade with  $k \geq x^{-1}$ .

**Proof.** Let  $c_1, c_2, c_3 > 0$ ,  $c_4, c_5 \geq 5$  be given by Lemma 4.3,  $c = 2^{-4c_5}$  and  $\xi = c^{10}$ . By Theorem 1.1 (with  $\epsilon = \xi$ ), there exists  $0 < \delta \leq 1$  such that every  $\overline{\mathcal{F}}$ -free graph has an  $\xi$ -restricted induced subgraph of size at least  $\delta|G|$ . Let  $d$  be sufficiently large so that  $2^{-d} < c^{10}$ ,  $\delta \geq 2^{58c_4-d}$  (which implies  $d \geq 58c_4$ ),  $\delta \geq 2^{c_2 + \frac{27c_4}{c_3} - d}$ . In the following we prove that  $d$  and  $c_1, c_4$  satisfy the lemma.

Fix an  $\overline{\mathcal{F}}$ -free graph  $G$  with  $|G| \geq x^{-d}$ . Since  $G$  is  $\overline{\mathcal{F}}$ -free,  $G$  has an  $\xi$ -restricted induced subgraph  $F$  of size at least  $\delta|G|$ . Since  $x \leq 2^{-d} < c^{10}$ , it follows from Lemma 4.3 with  $G$  replaced by  $F$  that one of the following holds.

- $F$  has an  $x$ -restricted induced subgraph  $S$  of size at least  $x^{22c_4}|F|$ .
- $F$  has a clique or stable set of size at least  $(x^{30c_4}|F|)^{c_1}$ .
- $F$  has a complete or anti-complete  $(k, \frac{|F|}{k^{c_2 + \frac{27c_4}{c_3}}})$ -blockade with  $k \geq 2$ .
- $F$  has an  $x$ -sparse or pure  $(\ell, |F|/\ell^{29c_4})$  blockade with  $\ell \in [c^{-1}, x^{-2}]$ .

If the first bullet holds, then the first outcome of Lemma 4.4 holds since  $x^{22c_4}|F| \geq x^{22c_4} \cdot \delta|G| \geq x^{22c_4} \cdot 2^{58c_4-d}|G| \geq x^{23c_4}|G|$ , where the last inequality is due to  $2^{-d} > x$ .

If the second bullet holds, then the third outcome of Lemma 4.4 holds since  $x^{30c_4}|F| \geq x^{30c_4} \cdot \delta|G| \geq x^{30c_4} \cdot 2^{58c_4-d}|G| \geq x^{31c_4}|G|$ .

If the third bullet holds, then the second or the fourth outcome of Lemma 4.4 holds since  $\frac{|F|}{k^{c_2+\frac{27c_4}{c_3}}} \geq \frac{\delta|G|}{k^{c_2+\frac{27c_4}{c_3}}} \geq \frac{|G|}{k^{c_2+\frac{27c_4}{c_3} \cdot 2^{d-(c_2+\frac{27c_4}{c_3})}}} \geq \frac{|G|}{k^d}$ , where the second inequality is due to  $\delta \geq 2^{c_2+\frac{27c_4}{c_3}-d}$ .

If the fourth bullet holds, then the second outcome holds with  $k = \ell^{1/2} \in [2, x^{-1}]$ , since  $\frac{|F|}{\ell^{29c_4}} \geq \frac{\delta|G|}{(\ell^{1/2})^{58c_4}} \geq \frac{|G|}{(\ell^{1/2})^{58c_4} \cdot 2^{d-58c_4}} \geq \frac{|G|}{(\ell^{1/2})^d}$ .

This completes the proof of Lemma 4.4.  $\square$

Now we are ready to prove Lemma 1.13, which we restate here for readers' convenience.

**Lemma 4.5.** *Suppose that  $\mathcal{F}$  has property  $(*)$  and  $\mathcal{F}$  is leaf-reducible. Then  $\mathcal{F}$  is generalized nice, that is, there exist constants  $c_1 \geq 3$ ,  $c_2 \geq 8$ ,  $c_3, c_4, c_5, c_8 > 0$ ,  $c_7 \geq 4$  and  $c_6 \geq 1$  such that for every  $\overline{\mathcal{F}}$ -free graph  $G$  and every  $\epsilon \in (0, \frac{1}{2})$ , either*

- (1)  $G$  has an  $(\epsilon^{-1}, \epsilon^{c_1}|G|)$ -blockade whose blocks are pairwise complete or weakly  $\epsilon^{c_2}$ -sparse;
- (2)  $G$  has a clique or stable set of size at least  $(\epsilon^{c_3}|G|)^{c_4}$ ;
- (3)  $G$  has a complete or anti-complete  $(k, |G|/k^{c_5})$ -blockade with  $k \geq \epsilon^{-c_6}$ ;
- (4)  $F$  has an  $\epsilon^{c_7}$ -restricted induced subgraph  $S$  of size at least  $\epsilon^{c_8}|G|$ .

**Proof.** Choose  $d \geq 58$  and  $c'_1 > 0, c'_4 \geq 4$  to satisfy Lemma 4.4 (with  $c_1$  replaced by  $c'_1$  and  $c_4$  replaced by  $c'_4$ ). We claim that  $c_1 = 10d^2, c_2 = d, c_3 = 156c'_4d, c_4 = c'_1, c_5 = 2d, c_6 = 5d, c_7 = 5d$  and  $c_8 = 116c'_4d$  suffice. Let  $x = \epsilon^{5d}$ . Suppose that none of the outcomes holds. If  $|G| \leq \epsilon^{-1}$ , then the second outcome of Lemma 4.5 holds. If  $\epsilon^{-1} \leq |G| \leq \epsilon^{-10d^2}$ , then the first outcome of Lemma 4.5 holds. So  $|G| \geq \epsilon^{-10d^2}$ . We then present a claim to give some property of an induced subgraph of  $G$  with at least  $\epsilon^d|G|$  vertices.

**Claim 4.5.1.** *Every induced subgraph  $F$  of  $G$  of size at least  $\epsilon^d|G|$  has a pure or  $x$ -sparse  $(k, |F|/k^d)$ -blockade with  $k \in [2, x^{-1}]$ .*

**Proof of Claim 4.5.1.** Since  $|G| \geq \epsilon^{-10d^2}$ ,  $|F| \geq \epsilon^d|G| \geq \epsilon^{-10d^2+d} \geq \epsilon^{-5d^2} = x^{-d}$ . Note that  $x \in (0, 2^{-d})$ . By Lemma 4.4, one of the following holds.

- $F$  has an  $x$ -restricted induced subgraph  $S$  of size at least  $x^{23c'_4}|F|$ .
- $F$  has a pure or  $x$ -sparse  $(k, |F|/k^d)$ -blockade with  $k \in [2, x^{-1}]$ .
- $F$  has a clique or stable set of size at least  $(x^{31c'_4}|F|)^{c_1}$ .
- $F$  has a complete or anti-complete  $(k, |F|/k^d)$ -blockade with  $k \geq x^{-1}$ .

If the first bullet holds, then the fourth outcome of Lemma 4.5 holds since  $x^{23c'_4}|F| \geq x^{23c'_4} \cdot \epsilon^d|G| \geq \epsilon^{116c'_4d}|G|$ .

If the third bullet holds, then the second outcome of Lemma 4.5 holds since  $x^{31c'_4}|F| \geq \epsilon^{155c'_4d} \cdot \epsilon^d|G| \geq \epsilon^{156c'_4d}|G|$ .

If the fourth bullet holds, then the third outcome of Lemma 4.5 holds since  $\frac{|F|}{k^d} \geq \frac{x^{1/5}|G|}{k^d} \geq \frac{k^{-1/5}|G|}{k^d} \geq \frac{|G|}{k^{2d}}$ .

So the second bullet holds, which proves Claim 4.5.1.  $\blacksquare$

By Claim 4.5.1 and Lemma 2.5,  $G$  has an  $(\epsilon^{-1}, \epsilon^{10d^2}|G|)$ -blockade whose blocks are pairwise complete or weakly  $\epsilon^d$ -sparse, which gives the first outcome of Lemma 4.5.

This completes the proof of Lemma 4.5.  $\square$

## 5 Deducing property (\*)

In this section, we prove Lemma 1.14 which we restate for reader's convenience.

**Lemma 5.1.** *Let  $\mathcal{F}_1$  and  $\mathcal{F}_2$  be two finite sets of graphs that satisfy the Erdős-Hajnal property. Let  $\mathcal{H}$  be a finite set of graphs. Suppose that for every  $\overline{\mathcal{H}}$ -free graph  $G$  and every  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$  with  $\ell, w \geq 4$ , if there exists a special vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ ,  $B_i$  (for each  $i \in [\ell]$ ) can be partitioned into  $X_i, Y_i$  such that*

- (1)  $Y_i$  is  $\mathcal{F}_1$ -free;
- (2)  $X_i$  can be partitioned into  $(A_1^i, \dots, A_{t_i}^i)$  such that
  - (2.1)  $(A_1^i, \dots, A_{t_i}^i)$  is a pure blockade;
  - (2.2) the pattern of  $(A_1^i, \dots, A_{t_i}^i)$ , whose vertex set consists of all blocks of  $(A_1^i, \dots, A_{t_i}^i)$  and two vertices are adjacent if and only if their corresponding blocks are complete to each other, is  $\mathcal{F}_2$ -free;
  - (2.3) for each  $j \in [t_i]$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $A_j^i$ .

Then  $\mathcal{H}$  satisfies property (\*).

**Proof.** We need to show that there exist  $c_1, c_2, c_3 > 0$  such that the following holds for every  $\overline{\mathcal{H}}$ -free graph  $G$ . If there is a  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$  with  $\ell, w \geq 4$ , and there exists  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ , then one of the following holds.

- (a)  $G$  has a clique or stable set of size at least  $w^{c_1}$ .
- (b)  $G$  has a complete or anti-complete  $(k, w/k^{c_2})$ -blockade with  $k \geq \ell^{c_3}$ .
- (c)  $G$  has a pure  $(\ell, w/\ell^2)$ -blockade.

Let  $c = \min\{c(\mathcal{F}_1), c(\mathcal{F}_2)\}$  where  $c(\mathcal{F}_i)$  is the Erdős-Hajnal constant for  $\mathcal{F}_i$ . We claim that  $c_1 = \frac{c}{2}$ ,  $c_2 = \frac{10}{c}$  and  $c_3 = \frac{c}{4}$  suffice. Suppose that none of the outcomes holds. Since  $|B_i| \geq w$  for each  $i \in [\ell]$ , either  $|X_i| \geq \frac{w}{2}$  or  $|Y_i| \geq \frac{w}{2}$ . Suppose first that  $|Y_i| \geq \frac{w}{2}$  for some  $i \in [\ell]$ . Since  $Y_i$  is  $\mathcal{F}_1$ -free,  $Y_i$  has a clique or stable set with at least  $(\frac{w}{2})^c$  vertices. Since  $w \geq 4$ ,  $(\frac{w}{2})^c \geq w^{\frac{c}{2}} = w^{c_1}$ . So (a) holds. Thus, we may assume that  $|X_i| \geq \frac{w}{2}$  for each  $i \in [\ell]$ .

**Claim 5.1.1.** *There exists some index  $i \in [\ell]$  such that every block of the pure blockade  $(A_1^i, \dots, A_{t_i}^i)$  has size at most  $\frac{w}{2\ell}$ .*

**Proof of Claim 5.1.1.** Suppose to the contrary that there exists a block of  $(A_1^i, \dots, A_{t_i}^i)$ , say  $A_1^i$ , with at least  $\frac{w}{2\ell}$  vertices for every  $i \in [\ell]$ . By (2.3),  $(A_1^1, \dots, A_1^\ell)$  is a pure  $(\ell, \frac{w}{2\ell})$ -blockade. Since  $\ell \geq 4$ ,  $2\ell \leq \ell^2$  and so  $(A_1^1, \dots, A_1^\ell)$  is also a pure  $(\ell, \frac{w}{\ell^2})$ -blockade. So (c) holds, a contradiction. ■

By Claim 5.1.1, there is an index  $i \in [\ell]$  such that  $|A_j^i| \leq \frac{w}{2\ell}$  for every  $j \in [t_i]$ . Since  $|X_i| \geq \frac{w}{2}$ ,  $t_i \geq \ell$ . Without loss of generality, we may assume that  $|A_1^i| \geq |A_2^i| \geq \dots \geq |A_{t_i}^i|$ . Let  $q = \lceil \log_{\ell^{1/2}} t_i \rceil$ . Let  $S_1 = \bigcup_{j \leq \ell^{1/2}} A_j^i$ , and

$$S_k = \left( \bigcup_{j \leq \min\{\ell^{k/2}, t_i\}} A_j^i \right) \setminus \left( \bigcup_{k' < k} S_{k'} \right) \text{ for } k = 2, \dots, q.$$

Then  $|X_i| = \sum_{k=1}^q |S_k|$ .

**Claim 5.1.2.** *For each  $k \in [q-1]$ , there exists a block of  $S_k$  of size less than  $\frac{w}{\ell^{5k/2}}$ .*

**Proof of Claim 5.1.2.** Suppose to the contrary that every block of  $S_k$  has at least  $\frac{w}{\ell^{5k/2}}$  vertices. Since  $\ell, w \geq 4$ ,  $\lfloor \ell^{k/2} \rfloor^c \geq \ell^{ck/4}$ . Since the pattern of  $(A_1^i, \dots, A_{t_i}^i)$ , whose vertex set consists of all blocks of  $(A_1^i, \dots, A_{t_i}^i)$  and two vertices are adjacent if and only if their corresponding blocks are complete to each other, is  $\mathcal{F}_2$ -free, there is a complete or anti-complete  $(\ell^{ck/4}, \frac{w}{\ell^{5k/2}})$ -blockade in  $\bigcup_{j \leq k} S_j$ . Since  $c_2 = \frac{10}{c}$ , this blockade is also a  $(\ell^{ck/4}, \frac{w}{(\ell^{ck/4})^{c_2}})$ -blockade. Then (b) holds, a contradiction. ■

By Claim 5.1.2, for each  $k \in [q-1]$ , there is a block in  $S_k$  of size less than  $\frac{w}{\ell^{5k/2}}$ . So every

block in  $S_{k+1}$  has at most  $\frac{w}{\ell^{5k/2}}$  vertices for each  $k \in [q-1]$ . Thus,

$$\begin{aligned}
|X_i| &= \sum_{k=1}^q |S_k| \\
&\leq |S_1| + \sum_{k=1}^{q-1} (\ell^{(k+1)/2}) \cdot \frac{w}{\ell^{5k/2}} \\
&\leq \ell^{1/2} \cdot \frac{w}{2\ell} + \sum_{k=1}^{q-1} (\ell^{(k+1)/2}) \cdot \frac{w}{\ell^{5k/2}} \\
&= w \left( \frac{1}{2\ell^{1/2}} + \sum_{k=1}^{q-1} \left( \frac{1}{\ell^{1/2}} \right)^{5k-k-1} \right) \\
&\leq w \left( \frac{1}{4} + \sum_{k=1}^{q-1} \left( \frac{1}{2} \right)^{k+2} \right) \\
&< \frac{w}{2}.
\end{aligned}$$

This contradicts the fact that  $|X_i| \geq \frac{w}{2}$ . This completes the proof of Lemma 5.1.  $\square$

**Remark.** If we set  $\mathcal{F}_2 = \{2K_1\}$ ,  $Y_i = \emptyset$  and partition  $X_i$  into anti-connected components, both  $P_5$  and bull satisfy the conditions of Lemma 5.1. Moreover, it is easy to see that each of  $P_5$  and bull is leaf-reducible and wonderful. This already proves that Erdős-Hajnal conjecture holds for  $P_5$  and bull.

## 6 Completing the proof

In this section, we prove that co- $E$ -free graphs and co-Bird-free graphs satisfy the hypothesis of Lemma 5.1, which implies that  $E$ -graph and Bird satisfy the Erdős-Hajnal property. We first define the quotient blockade of a blockade and derive some useful properties of quotient blockades.

Suppose that  $\mathcal{L} = (L_1, \dots, L_n)$  is a blockade. We say  $L_i$  and  $L_j$  have *relation*  $\mathcal{M}$  if and only if  $i = j$  or there is a block sequence  $L_i = L_{r_1}, L_{r_2}, \dots, L_{r_m} = L_j$  such that for each  $k \in [m-1]$ ,  $L_{r_k}$  and  $L_{r_{k+1}}$  are mixed. It is easy to check that  $\mathcal{M}$  is an equivalence relation. Let  $\mathcal{L}/\mathcal{M}$  be the *quotient blockade* of  $\mathcal{L}$  where each block of  $\mathcal{L}/\mathcal{M}$  is the union of all blocks in an equivalence class of  $\mathcal{M}$ . We present two properties of the quotient blockade.

**Lemma 6.1.** *Suppose that  $\mathcal{L}$  is a blockade such that each block of  $\mathcal{L}$  is connected (anti-connected). Then*

- (1) *Each block of  $\mathcal{L}/\mathcal{M}$  is also connected (resp. anti-connected).*
- (2) *Each two blocks of  $\mathcal{L}$  that are contained in different blocks of  $\mathcal{L}/\mathcal{M}$  are pure to each other.*
- (3) *If there are a block  $D$  of  $\mathcal{L}/\mathcal{M}$  and a vertex  $u \notin D$  that is mixed on  $D$  but is pure to each block of  $\mathcal{L}$  contained in  $D$ , then there are two mixed blocks of  $\mathcal{L}$  contained in  $D$  such that  $u$  is complete to one of the blocks and anti-complete to the other.*

**Proof.** The first two statements of Lemma 6.1 follow immediately from the definition of  $\mathcal{L}/\mathcal{M}$ . We now prove (3). Since  $u$  is mixed on  $D$  but is pure to each block of  $\mathcal{L}$  contained in  $D$ , there are two blocks of  $\mathcal{L}$ , say  $A_1, A_2$ , that are contained in  $D$  such that  $u$  is complete to  $A_1$  and anti-complete to  $A_2$ . Since  $D$  is an equivalence class of  $\mathcal{L}$  with respect to  $\mathcal{M}$ , there is a block sequence  $A_1 = A_{r_1}, A_{r_2}, \dots, A_{r_m} = A_2$  such that for each  $k \in [m-1]$ ,  $A_{r_k}$  and  $A_{r_{k+1}}$  are mixed. So there is an index  $k$  such that  $u$  is complete to  $A_{r_k}$  and anti-complete to  $A_{r_{k+1}}$ . This proves Lemma 6.1 (3).  $\square$

**Lemma 6.2.** *Suppose that  $\mathcal{L}$  is a blockade in  $G$  such that each block of  $\mathcal{L}$  is connected (anti-connected). If there are two mixed blocks  $D_1, D_2$  of  $\mathcal{L}/\mathcal{M}$  and three vertices  $x, y, u \notin D_1 \cup D_2$  such that*

- $x, y$  are two non-adjacent vertices that are complete to  $D_1 \cup D_2$ ,
- $u \in N(x) \setminus N(y)$  is complete to  $D_1$  but is anti-complete to  $D_2$ , and
- no vertex in  $D_1$  can mix on  $D_2$ ,

*then there are two mixed blocks  $A_1, A_2$  of  $\mathcal{L}$  that contained in  $D_1$  and three vertices  $x', y', u' \notin A_1 \cup A_2$  such that*

- (1)  $x', y'$  are two non-adjacent vertices that are complete to  $A_1 \cup A_2$  and
- (2)  $u' \in N(x') \setminus N(y')$  is complete to  $A_1$  but is anti-complete to  $A_2$ .

**Proof.** Since  $D_1$  and  $D_2$  are mixed, either there is a vertex  $b_1 \in D_1$  mixed on  $D_2$  or there is a vertex  $b_2 \in D_2$  mixed on  $D_1$ . By our assumption, there is a vertex  $b_2 \in D_2$  mixed on  $D_1$ . By Lemma 6.1 (2) and (3), there are two mixed blocks of  $\mathcal{L}$ , say  $A_1, A_2$ , contained in  $D_1$  such that  $b_2$  is complete to  $A_1$  but is anti-complete to  $A_2$ . Then  $(x', y', u') = (y, u, b_2)$  are the desired vertices.  $\square$

## 6.1 E-graph

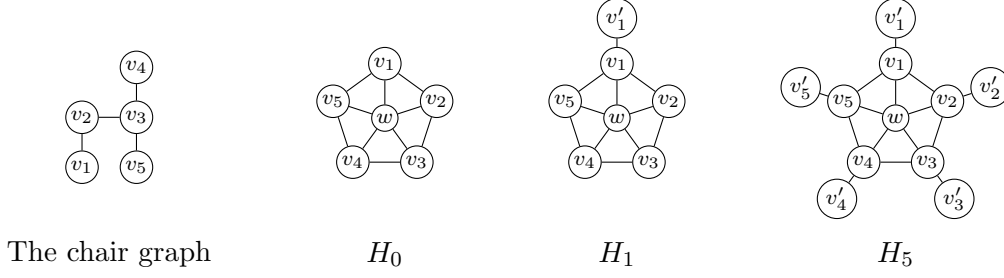
In this subsection, we prove that co- $E$ -graph-free graphs satisfy the hypothesis of Lemma 5.1.

Let *chair* be the 5-vertex graph obtained from an induced  $P_4$  by adding a new vertex pendent to the second vertex of the  $P_4$  and *co-chair* be the complement of the chair graph. Let  $H_0$  be a 5-wheel labeled as in Figure 7. For each integer  $1 \leq i \leq 5$ , let  $H_i$  be the graph obtained from  $H_{i-1}$  by adding a vertex  $v'_i$  and an edge  $v_i v'_i$ , see Figure 7 for an illustration of chair,  $H_0, H_1$  and  $H_5$ .

Let us first record the following simple lemma.

**Lemma 6.3.**  $\{H_5, \text{co-}E\}$  has the Erdős-Hajnal property.

**Proof.** By Theorems 1.4 and 1.5,  $H_0$  and co-chair have the Erdős-Hajnal property. It follows that  $\{H_0, \text{co-}E\}$  and  $\{H_i, \text{co-chair}\}$  have the Erdős-Hajnal property for every  $i \in [5]$ . Assume that  $\{H_{i-1}, \text{co-}E\}$  has the Erdős-Hajnal property for  $1 \leq i \leq 5$ . By Corollary 1.8 with  $F_1 = H_i$



**Fig. 7.** The chair graph,  $H_0$ ,  $H_1$  and  $H_5$ .

and  $F_2$  being the  $E$ -graph,  $\{H_i, \text{co-}E\}$  has the Erdős-Hajnal property. So  $\{H_5, \text{co-}E\}$  has the Erdős-Hajnal property.  $\square$

**Lemma 6.4.** *Let  $G$  be a  $\text{co-}E$ -free graph with a  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$  and a vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ . For each  $i \in [\ell]$ ,  $B_i$  can be partitioned into  $X_i, Y_i$  such that*

- (1)  $Y_i$  is  $\{H_5, \text{co-}E\}$ -free;
- (2)  $X_i$  can be partitioned into  $(A_1^i, \dots, A_{t_i}^i)$  such that
  - (2.1)  $(A_1^i, \dots, A_{t_i}^i)$  is a pure blockade;
  - (2.2) the pattern of  $(A_1^i, \dots, A_{t_i}^i)$ , whose vertex set consists of all blocks of  $(A_1^i, \dots, A_{t_i}^i)$  and two vertices are adjacent if and only if their corresponding blocks are complete to each other, is  $\{H_5, \text{co-}E\}$ -free;
  - (2.3) for each  $j \in [t_i]$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $A_j^i$ .

**Proof.** We begin with a useful claim.

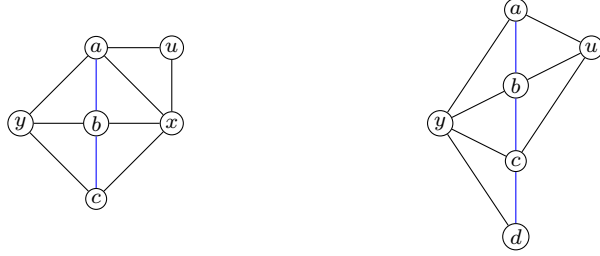
**Claim 6.4.1.** *Let  $P$  be an induced path in  $G$  and  $x, y \in V(G) \setminus V(P)$  be two non-adjacent vertices complete to  $P$ . For each vertex  $u \in N(x) \setminus N(y)$  that is mixed on  $P$ ,*

- (1)  $u$  has no two consecutive non-neighbors in  $P$ ; and
- (2)  $u$  has no three consecutive neighbors in  $P$ .

**Proof of Claim 6.4.1.** Let  $u \in N(x) \setminus N(y)$  be a vertex that is mixed on  $P$ . Suppose first that  $u$  has two consecutive non-neighbors in  $P$ . So there is an induced subpath  $a - b - c$  of  $P$  such that  $u$  is adjacent to  $a$  but is not adjacent to  $b, c$ . Then  $\{x, y, u, a, b, c\}$  induces a  $\text{co-}E$  (see Figure 8), a contradiction. This proves Claim 6.4.1 (1).

Now we suppose that  $u$  has three consecutive neighbors in  $P$ . So there is an induced subpath  $a - b - c - d$  of  $P$  such that  $u$  is adjacent to  $a, b, c$  but is not adjacent to  $d$ . Then  $\{y, u, a, b, c, d\}$  induces a  $\text{co-}E$  (see Figure 8), a contradiction. This proves Claim 6.4.1 (2).  $\blacksquare$

By Claim 6.4.1, we obtain a “homogeneous structure” in  $B_i$ . Formally, we have the following.



Proof of Claim 6.4.1 (1)

Proof of Claim 6.4.1 (2)

**Fig. 8.** Induced co- $E$ s in the proof of Claim 6.4.1 where the blue line is the subpath of  $P$ .

**Claim 6.4.2.** *If  $x, y$  are two non-adjacent vertices complete to an induced  $H_5$ , then every vertex  $u \in N(x) \setminus N(y)$  is pure to this  $H_5$ . In particular, for every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to every induced  $H_5$  in  $B_i$ .*

**Proof of Claim 6.4.2.** Suppose to the contrary that there is a vertex  $u \in N(x) \setminus N(y)$  that is mixed on an induced  $H_5$  labeled as in Figure 7. Let  $C := v_1 - v_2 - v_3 - v_4 - v_5 - v_1$ . All indices below are modulo 5.

Suppose first that  $u$  is complete to  $C$ . By Claim 6.4.1 (2) with  $P = v'_j - v_j - v_{j+1} - v_{j+2}$ , we have  $u$  is adjacent to  $v'_j$  for each  $j \in [5]$ . If  $u$  is not adjacent to  $w$ , then  $\{v'_1, u, v_1, v_2, v_3, w\}$  induces a co- $E$ , a contradiction. So we may assume  $uw \in E(G)$ . Then  $u$  is complete to  $H_5$ , a contradiction. So  $u$  is not complete to  $C$ .

Now we suppose that  $u$  is anti-complete to  $C$ . By Claim 6.4.1 (1) with  $P = v'_j - v_j - v_{j+1}$ , we have  $u$  is not adjacent to  $v'_j$  for each  $j \in [5]$ . By Claim 6.4.1 (1) with  $P = v'_1 - v_1 - w$ , we have  $u$  is not adjacent to  $w$ . It follows that  $u$  is anti-complete to  $H_5$ , a contradiction. So  $u$  is not anti-complete to  $C$ .

Therefore, we may assume  $u$  is mixed on  $C$ . By Claim 6.4.1 (1) and (2), there is an index  $j \in [5]$  such that  $N(u) \cap V(C) = \{v_j, v_{j-2}, v_{j+2}\}$ , say  $\{v_1, v_3, v_4\}$ . By Claim 6.4.1 (1) with  $P = v_4 - w - v_2$ , we have  $u$  is adjacent to  $w$ . By Claim 6.4.1 (2) with  $P = v'_1 - v_1 - w - v_3$ , we have  $u$  is adjacent to  $v'_1$ . Then  $\{u, w, v'_1, v_1, v_2, v_3\}$  induces a co- $E$ , a contradiction. This proves the first statement of the claim.

The second statement of Claim 6.4.2 follows immediately with  $(x, y)$  replaced by  $(v, a_i)$ . This completes the proof of Claim 6.4.2.  $\blacksquare$

Now we are ready to define  $X_i, Y_i$  and  $(A_1^i, \dots, A_{t_i}^i)$ . Let  $X_i \subseteq B_i$  be the set of all vertices contained in some induced  $H_5$  in  $B_i$  and  $Y_i = B_i \setminus X_i$ . So  $Y_i$  is  $H_5$ -free. Since  $G$  is co- $E$ -free,  $Y_i$  is  $\{H_5, \text{co-}E\}$ -free. This proves Lemma 6.4 (1). For any two vertices  $d, d' \in X_i$ , we say  $d, d'$  have relation  $\mathcal{R}$  if and only if there is a vertex sequence  $d = d_1, d_2, \dots, d_n = d'$  such that for each  $k \in [n-1]$ ,  $d_k$  and  $d_{k+1}$  are contained in the same induced  $H_5$  in  $B_i$ . It is easy to check that  $\mathcal{R}$  is an equivalence relation on  $X_i$ . Let  $\mathcal{L}^1$  be the blockade whose blocks are equivalence classes of  $\mathcal{R}$ . Since  $H_5$  is connected, each block of  $\mathcal{L}^1$  is connected. For each  $s \geq 2$ , let  $\mathcal{L}^s = \mathcal{L}^{s-1}/\mathcal{M}$ . Note that  $\mathcal{L}^s$  is different from  $\mathcal{L}^{s-1}$  if and only if there are two mixed blocks in  $\mathcal{L}^{s-1}$ . Since  $X_i$  is finite, there is an integer  $q$  such that  $\mathcal{L}^q$  is a pure blockade. Set  $(A_1^i, \dots, A_{t_i}^i) := \mathcal{L}^q$  and this

proves Lemma 6.4 (2.1). If there is an induced  $H_5$  such that each vertex of this  $H_5$  is contained in a different block of  $(A_1^i, \dots, A_{t_i}^i)$ , then these vertices should have been in the same block of  $\mathcal{L}^1$ , a contradiction. Since  $G$  is co- $E$ -free, the pattenr graph is  $\{H_5, \text{co-}E\}$ -free. This proves Lemma 6.4 (2.2). It remains to prove Lemma 6.4 (2.3).

**Claim 6.4.3.** *For every  $s \in [q]$ , every block  $L$  of  $\mathcal{L}^s$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $L$ .*

**Proof of Claim 6.4.3.** We prove this claim by induction on  $s$ . If  $s = 1$ , then this claim follows immediately from Claim 6.4.2 with  $(x, y, u) = (v, a_i, u)$ . So we may assume that  $s \geq 2$  and this claim holds for  $s - 1$ .

Suppose to the contrary that there is a vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$  that is mixed on  $L$ . By the inductive hypothesis,  $u$  is pure to each block of  $\mathcal{L}^{s-1}$ . By Lemma 6.1 (3), there are two mixed blocks of  $\mathcal{L}^{s-1}$  contained in  $L$  such that  $u$  is complete to one of these two blocks but is anti-complete to the other one. Suppose that for each fixed  $i \in [s]$ , there are two mixed blocks  $D_1, D_2$  of  $\mathcal{L}^i$  and three vertices  $x, y, z$  with  $z \in N(x) \setminus N(y)$  such that  $x, y$  are complete to  $D_1 \cup D_2$  and  $z$  is complete to  $D_1$  and anti-complete to  $D_2$ . If  $b_1 \in D_1$  mixes on  $D_2$ , then  $b_1$  mixed on an edge  $b_2 b'_2 \in D_2$  by connectivity. By Claim 6.4.1 (1) with  $(P, x, y, z) = (b_1 - b_2 - b'_2, x, y, z)$ , we obtain a contradiction since  $z$  is mixed on  $b_1 - b_2 - b'_2$  and has two consecutive non-neighbors  $b_2, b'_2$ .

By applying Lemma 6.2 repeatedly, beginning with  $(x, y, u) = (v, a_i, u)$ , where each application of Lemma 6.2 preserves the same hypotheses until reaching  $\mathcal{L}^1$ , there are two mixed blocks  $A_1, A_2$  of  $\mathcal{L}^1$  and three vertices  $x', y', u' \notin A_1 \cup A_2$  such that

- $x', y'$  are two non-adjacent vertices that are complete to  $A_1 \cup A_2$ ;
- $u' \in N(x') \setminus N(y')$  is complete to  $A_1$  but is anti-complete to  $A_2$ .

So we may assume that there is a vertex  $u_2 \in A_2$  mixed on  $A_1$ . By Claim 6.4.2 with  $(x, y, u) = (y', u', u_2)$ ,  $u_2$  is pure to every induced  $H_5$  of  $A_1$ . Since  $A_1$  is an equivalence class of  $\mathcal{R}$ ,  $u_2$  is pure to  $A_1$ , a contradiction.

This complete the proof of Claim 6.4.3. ■

By Claim 6.4.3 with  $s = q$ , Lemma 6.4 (2.3) holds. This completes the proof of Lemma 6.4. □

## 6.2 Bird

In this subsection, we prove that Bird-free graph satisfies the hypothesis of Lemma 5.1.

**Lemma 6.5.** *Let  $G$  be a co-Bird-free graph with a  $(\ell, w)$ -comb  $((a_i, B_i), i \in [\ell])$  in  $G$  and a vertex  $v \in V(G) \setminus (\{a_i : i \in [\ell]\} \cup \bigcup_{i \in [\ell]} B_i)$  such that  $v$  is complete to  $\bigcup_{i \in [\ell]} B_i$  and anti-complete to  $\{a_i : i \in [\ell]\}$ . For each  $i \in [\ell]$ ,  $B_i$  can be partitioned into  $X_i, Y_i$  such that*

- (1)  $Y_i$  is  $E$ -graph-free;

(2)  $X_i$  can be partitioned into  $(A_1^i, \dots, A_{t_i}^i)$  such that

(2.1)  $(A_1^i, \dots, A_{t_i}^i)$  is a pure blockade;

(2.2) the pattern of  $(A_1^i, \dots, A_{t_i}^i)$ , whose vertex set consists of all blocks of  $(A_1^i, \dots, A_{t_i}^i)$  and two vertices are adjacent if and only if their corresponding blocks are complete to each other, is  $E$ -graph-free;

(2.3) for each  $j \in [t_i]$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $A_j^i$ .

**Proof.** We begin with a useful claim.

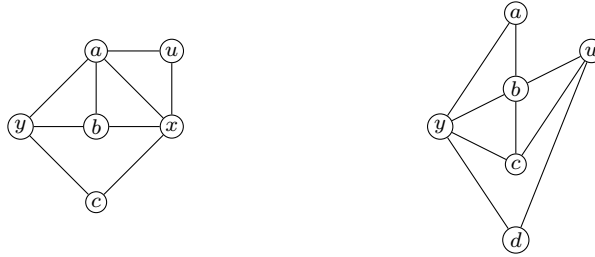
**Claim 6.5.1.** Let  $S$  be an induced subgraph of  $G$  and  $x, y \in V(G) \setminus V(S)$  be two non-adjacent vertices complete to  $S$ . For every induced subgraph  $H$  of  $S$ , there is no  $u \in N(x) \setminus N(y)$  for which either of the following holds.

(1)  $H$  is  $P_2 + P_1$  and  $u$  is mixed on the  $P_2$  but is anti-complete to the  $P_1$ .

(2)  $H$  is  $P_3 + P_1$  and  $u$  is adjacent to the  $P_1$  and two consecutive vertices of the  $P_3$  but is anti-complete to the remaining vertex of the  $P_3$ .

**Proof of Claim 6.5.1.** Let  $u \in N(x) \setminus N(y)$  be a vertex. Suppose first that  $H$  is  $P_2 + P_1$  with edge  $ab$  and the isolated vertex  $c$ . If  $u$  is adjacent to  $a$  but is non-adjacent to  $b$  and  $c$ , then  $\{x, y, u, a, b, c\}$  induces a co-Bird (see Figure 9), a contradiction. This proves Claim 6.5.1 (1).

Now we suppose that  $H$  is  $P_3 + P_1$  with induced path  $a - b - c$  and the isolated vertex  $d$ . If  $u$  is adjacent to  $b, c, d$  but is non-adjacent to  $a$ , then  $\{y, u, a, b, c, d\}$  induces a co-Bird (see Figure 9), a contradiction. This proves Claim 6.5.1 (2). ■



Proof of Claim 6.5.1 (1)

Proof of Claim 6.5.1 (2)

**Fig. 9.** Induced co-Birds in the proof of Claim 6.5.1.

**Claim 6.5.2.** If  $x, y$  are two non-adjacent vertices complete to an induced  $E$ -graph, then every vertex  $u \in N(x) \setminus N(y)$  is pure to this  $E$ -graph. In particular, for every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to every induced  $E$ -graph in  $B_i$ .

**Proof of Claim 6.5.2.** Suppose to the contrary that there is a vertex  $u \in N(x) \setminus N(y)$  that is mixed on an induced  $E$ -graph. Let  $P = v_1 - v_2 - v_3 - v_4 - v_5$  be the induced  $P_5$  contained in this  $E$ -graph and  $v'_3$  be the vertex adjacent to  $v_3$ .

Suppose first that  $u$  is complete to  $P$ . By Claim 6.5.1 (2) with  $H$  induced by  $\{v'_3, v_3, v_4, v_1\}$ , we have  $u$  is adjacent to  $v'_3$ . Then  $u$  is complete to this  $E$ -graph, a contradiction. So  $u$  is not complete to  $P$ . Now we suppose that  $u$  is anti-complete to  $P$ . By Claim 6.5.1 (1) with  $H$  induced by  $\{v_1, v_3, v'_3\}$ , we have  $u$  is not adjacent to  $v'_3$ . It follows that  $u$  is anti-complete to this  $E$ -graph, a contradiction. So  $u$  is not anti-complete to  $P$ .

Therefore, we may assume  $u$  is mixed on  $P$ . Suppose first that  $u$  is mixed on  $v_1v_2$ . By Claim 6.5.1 (1),  $u$  is adjacent to each of  $v'_3, v_4, v_5$ . If  $u$  is adjacent to  $v_1$  but is not adjacent to  $v_2$ , then  $u$  is adjacent to  $v_3$  by Claim 6.5.1 (2) with  $H$  induced by  $\{v_1, v_3, v_4, v_5\}$ . Then the  $\{v_2, v_3, v'_3, v_5\}$  contradicts Claim 6.5.1 (2). So we may assume that  $u$  is adjacent to  $v_2$  but is not adjacent to  $v_1$ . By Claim 6.5.1 (1) with  $H$  induced by  $\{v_1, v_3, v'_3\}$ ,  $u$  is adjacent to  $v_3$ . Then  $\{v_1, v_2, v_3, v_5\}$  contradicts Claim 6.5.1 (2). This proves that  $u$  is pure to  $v_1v_2$ . By symmetry,  $u$  is pure to  $v_4v_5$ .

If  $u$  is adjacent to  $v_1, v_2, v_4, v_5$ , then  $u$  is non-adjacent to  $v_3$  and then  $\{v_1, v_2, v_3, v_5\}$  contradicts Claim 6.5.1 (2). If  $u$  is non-adjacent to  $v_1, v_2, v_4, v_5$ , then  $u$  is adjacent to  $v_3$  and so  $\{v_2, v_3, v_5\}$  contradicts Claim 6.5.1 (1). So we may assume that  $u$  is adjacent to  $v_1, v_2$  but is non-adjacent to  $v_4, v_5$ . By Claim 6.5.1 (1) with  $H$  induced by  $\{v_2, v_3, v_5\}$ ,  $u$  is adjacent to  $v_3$ . By Claim 6.5.1 (2) with  $H$  induced by  $\{v_1, v'_3, v_3, v_4\}$ ,  $u$  is not adjacent to  $v'_3$ , which gives a contradiction by Claim 6.5.1 (1) with  $H$  induced by  $\{v_3, v'_3, v_5\}$ . This proves the first statement of the claim.

The second statement of Claim 6.5.2 follows immediately with  $(x, y)$  replaced by  $(v, a_i)$ . This completes the proof of Claim 6.5.2. ■

Now we are ready to define  $X_i, Y_i$  and  $(A_1^i, \dots, A_{t_i}^i)$ . Let  $X_i \subseteq B_i$  be the set of all vertices contained in some induced  $E$ -graph in  $B_i$  and  $Y_i = B_i \setminus X_i$ . So  $Y_i$  is  $E$ -graph-free. This proves Lemma 6.5 (1). For any two vertices  $d, d' \in X_i$ , we say  $d, d'$  have *relation*  $\mathcal{R}$  if and only if there is a vertex sequence  $d = d_1, d_2, \dots, d_n = d'$  such that for each  $k \in [n-1]$ ,  $d_k$  and  $d_{k+1}$  are contained in the same induced  $E$ -graph in  $B_i$ . It is easy to check that  $\mathcal{R}$  is an equivalence relation on  $X_i$ . Let  $\mathcal{L}^1$  be the blockade whose blocks are equivalence classes of  $\mathcal{R}$ . Since  $E$ -graph is anti-connected, each block of  $\mathcal{L}^1$  is anti-connected. For each  $s \geq 2$ , let  $\mathcal{L}^s = \mathcal{L}^{s-1}/\mathcal{M}$ . Note that  $\mathcal{L}^s$  is different from  $\mathcal{L}^{s-1}$  if and only if there are two mixed blocks in  $\mathcal{L}^{s-1}$ . Since  $X_i$  is finite, there is an integer  $q$  such that  $\mathcal{L}^q$  is a pure blockade. Set  $(A_1^i, \dots, A_{t_i}^i) := \mathcal{L}^q$  and this proves Lemma 6.5 (2.1). If there is an induced  $E$ -graph such that each vertex of this  $E$ -graph is contained in a different block of  $(A_1^i, \dots, A_{t_i}^i)$ , then these vertices should have been in the same block of  $\mathcal{L}^1$ , a contradiction. This proves Lemma 6.5 (2.2). It remains to prove Lemma 6.5 (2.3).

**Claim 6.5.3.** *For every  $s \in [q]$ , every block  $L$  of  $\mathcal{L}^s$  and every vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$ ,  $u$  is pure to  $L$ .*

**Proof of Claim 6.5.3.** We prove this claim by induction on  $s$ . If  $s = 1$ , then this claim follows immediately from Claim 6.5.2 with  $(x, y, u) = (v, a_i, u)$ . So we may assume that  $s \geq 2$  and this claim holds for  $s - 1$ .

Suppose to the contrary that there is a vertex  $u \in \bigcup_{k \in [\ell] \setminus \{i\}} B_k$  that is mixed on  $L$ . By the

inductive hypothesis,  $u$  is pure to each block of  $\mathcal{L}^{s-1}$ . By Lemma 6.1 (3), there are two mixed blocks of  $\mathcal{L}^{s-1}$  contained in  $L$  such that  $u$  is complete to one of these two blocks but is anti-complete to the other one. Suppose that for each fixed  $i \in [s]$ , there are two mixed blocks  $D_1, D_2$  of  $\mathcal{L}^i$  and three vertices  $x, y, z$  with  $z \in N(x) \setminus N(y)$  such that  $x, y$  are complete to  $D_1 \cup D_2$  and  $z$  is complete to  $D_1$  and anti-complete to  $D_2$ . If  $b_1 \in D_1$  mixes on  $D_2$ , then  $b_1$  mixed on a non-edge  $b_2 b'_2 \in D_2$  by anti-connectivity. By Claim 6.5.1 (1) with  $(H, x, y, z) = (\{b_1 b_2\} + b'_2, x, y, z)$ , we obtain a contradiction.

By applying Lemma 6.2 repeatedly, beginning with  $(x, y, u) = (v, a_i, u)$ , where each application of Lemma 6.2 preserves the same hypotheses until reaching  $\mathcal{L}^1$ , there are two mixed blocks  $A_1, A_2$  of  $\mathcal{L}^1$  and three vertices  $x', y', u' \notin A_1 \cup A_2$  such that

- $x', y'$  are two non-adjacent vertices that are complete to  $A_1 \cup A_2$ , and
- $u' \in N(x') \setminus N(y')$  is complete to  $A_1$  but is anti-complete to  $A_2$ .

So we may assume that there is a vertex  $u_2 \in A_2$  mixed on  $A_1$ . By Claim 6.5.2 with  $(x, y, u) = (y', u', u_2)$ ,  $u_2$  is pure to every induced  $E$ -graph of  $A_1$ . Since  $A_1$  is an equivalence class of  $\mathcal{R}$ ,  $u_2$  is pure to  $A_1$ , a contradiction.

This complete the proof of Claim 6.5.3. ■

By Claim 6.5.3 with  $s = q$ , Lemma 6.5 (2.3) holds. This completes the proof of Lemma 6.5. □

## 7 Concluding remarks

In this paper, we combine several new ideas to extend the iterative sparsification framework recently developed by Nguyen, Scott and Seymour [15]. Our framework not only yields the first two six-vertex graphs satisfying the Erdős-Hajnal conjecture which do not follow directly from Theorem 1.4 and Corollary 1.8, but also provides a general criterion (Lemma 1.14) that may be used to verify the conjecture for further graph classes. Moreover, we recover the result of Chudnovsky and Safra [6] on bull graph and the result of Nguyen, Scott and Seymour [15] on  $P_5$  in a unified way. This suggests that the comb-based approach is a promising direction for future work on the Erdős-Hajnal conjecture.

## Acknowledgments

We thank Viet Hoang and his coauthors for pointing out a gap in an earlier version of our proof of Lemma 2.8, and for their helpful comments.

## References

- [1] N. Alon, J. Pach, and J. Solymosi. Ramsey-type theorems with forbidden subgraphs. *Combinatorica*, 21(2):155-170, 2001.

- [2] J. A. Bondy and U. S. R. Murty. *Graph Theory*. Springer, 2008.
- [3] N. Bousquet, A. Lagoutte, and S. Thomassé. The Erdős–Hajnal conjecture for paths and antipaths. *Journal of Combinatorial Theory, Series B*, 113:261-264, 2015.
- [4] M. Bucić, J. Fox, and H. T. Pham. Equivalence between Erdős-Hajnal and polynomial Rödl and Nikiforov conjectures. arXiv:2403.08303 [math.CO], 2024.
- [5] M. Bucić, T. Nguyen, A. Scott and P. Seymour. A log log step towards Erdős-Hajnal. *International Mathematics Research Notices*, 9991-10004, 2024.
- [6] M. Chudnovsky and S. Safra. The Erdős-Hajnal conjecture for bull-free graphs. *Journal of Combinatorial Theory, Series B*, 98(6):1301-1310, 2008.
- [7] M. Chudnovsky, A. Scott, P. Seymour, and S. Spirkl. Pure pairs. I. Trees and linear anti-complete pairs. *Advances in Mathematics*, 375:107396, 20, 2020.
- [8] M. Chudnovsky, A. Scott, P. Seymour, and S. Spirkl. Erdős-Hajnal for graphs with no 5-hole. *Proceedings of the London Mathematical Society*, 126(3):997-1014, 2023.
- [9] P. Erdős. Some remarks on the theory of graphs. *Bulletin of the American Mathematical Society*, 53:292-294, 1947.
- [10] P. Erdős and A. Hajnal. Ramsey-type theorems. *Discrete Applied Mathematics*, 25:37-52, 1989.
- [11] P. Erdős and G. Szekeres. A combinatorial problem in geometry. *Compositio Mathematica*, 2:464-470, 1935.
- [12] J. Fox and B. Sudakov. Induced Ramsey-type theorems. *Advances in Mathematics*, 219:1771–1800, 2008.
- [13] T. Nguyen, A. Scott, and P. Seymour. Induced subgraphs density. IV. New graphs with the Erdős-Hajnal property. *Transactions of the American Mathematical Society*, accepted, 2026.
- [14] T. Nguyen, A. Scott, and P. Seymour. Induced subgraph density. VI. Bounded VC-dimension. *Advances in Mathematics*, 482, no. A, 110601, 2025.
- [15] T. Nguyen, A. Scott, and P. Seymour. Induced subgraph density. VII. The five-vertex path. *Proceedings of the London Mathematical Society*, 132(3):e70133, 2026.
- [16] V. Nikiforov. Edge distribution of graphs with few copies of a given graph. *Combinatorics, Probability and Computing*, 15(6):895-902, 2006.
- [17] F. P. Ramsey. On a problem of formal logic. *Proceedings of the London Mathematical Society*, 30:264-286, 1930.
- [18] V. Rödl. On universality of graphs with uniformly distributed edges. *Discrete Mathematics*, 59(1-2):125-134, 1986.